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371.

ON A FORMULA FOR THE INTERSECTIONS OF A LINE AND CONIC, AND ON AN INTEGRAL FORMULA CONNECTED THEREWITH.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vii. (1866), pp. 1–6.]

In a letter to me, dated 15 May, 1862, Mr Spottiswoode has extracted from an unpublished Memoir, and he has kindly permitted me to communicate, the following formula for the points of intersection of a line and conic; viz. if the equations of the line and conic are

$$\xi x + \eta y + \zeta z = 0, \quad (a, b, c, f, g, h \mid x, y, z)^2 = 0;$$

and if

$$\theta = - \begin{vmatrix} \cdot & \xi & \eta & \zeta \\ \xi & a & h & g \\ \eta & h & b & f \\ \zeta & g & f & c \end{vmatrix},$$

or, what is the same thing, if

$$\theta^2 = -(A, B, C, F, G, H \mid \xi, \eta, \zeta)^2,$$

where $A = bc - f^2$, etc. as usual; then the coordinates (x, y, z) of a point of intersection of the line and conic are found from the linear equations

$$\begin{aligned} (g\eta - h\zeta - \theta)x + (f\eta - b\zeta)y + (c\eta - f\zeta)z &= 0, \\ (a\zeta - g\xi)x + (h\zeta - f\xi - \theta)y + (g\zeta - c\xi)z &= 0, \\ (h\xi - a\eta)x + (b\xi - h\eta)y + (f\xi - g\eta - \theta)z &= 0, \end{aligned}$$

equivalent of course to two equations, and giving by the elimination of (x, y, z) , the equation

$$\theta [-(A, B, C, F, G, H \mid \xi, \eta, \zeta)^2 - \theta^2] = 0,$$

that is, giving for θ the foregoing value. And the linear equations then give

$$x : y : z = \theta \frac{d\theta}{d\xi} + g\eta - h\zeta : \theta \frac{d\theta}{d\eta} + f\zeta - h\xi : \theta \frac{d\theta}{d\zeta} + h\xi - f\eta,$$

where obviously

$$-\theta \frac{d\theta}{d\xi} = A\xi + H\eta + G\zeta, \quad -\theta \frac{d\theta}{d\eta} = H\xi + B\eta + F\zeta, \quad -\theta \frac{d\theta}{d\zeta} = G\xi + F\eta + C\zeta.$$

By changing the sign of θ , we have of course the coordinates of the other point of intersection. The formulae which, singularly enough, have since been given incidentally by M. Aronhold¹, may be easily obtained as follows.

Writing for shortness

$$P = ax + hy + gz, \quad Q = hx + by + fz, \quad R = gx + fy + cz,$$

then the equation of the conic gives

$$Px + Qy + Rz = 0,$$

and combining with this the equation

$$\xi x + \eta y + \zeta z = 0,$$

¹In his interesting Memoir “Ueber eine neue algebraische Behandlungsweise der Integrale irrationaler Differentiale von der Form $\Pi(x, y) dx$, in welcher $\Pi(x, y)$ eine beliebige rationale Function ist, und zwischen x und y eine allgemeine Gleichung zweiter Ordnung besteht.” *Crelle*, t. lxi. (1862).

we have

$$x : y : z = Q\zeta - R\eta : R\xi - P\zeta : P\eta - Q\xi,$$

or what is the same thing, taking an indeterminate multiplier θ ,

$$\begin{aligned} -\theta x + R\eta - Q\zeta &= 0, \\ -\theta y + P\zeta - R\xi &= 0, \\ -\theta z + Q\xi - P\eta &= 0, \end{aligned}$$

which are, in fact, Mr Spottiswoode's linear equations, and which lead, as before, to the value of θ .

But this value of θ is obtained in a different manner by expressing (x, y, z) as linear functions of P, Q, R ; viz. putting as usual $K = abc - af^2 - bg^2 - ch^2 + 2fgh$, the linear equations thus become

$$\begin{aligned} AP + HQ + GR + \frac{K}{\theta}(\eta R - \zeta Q) &= 0, \\ HP + BQ + FR + \frac{K}{\theta}(\zeta P - \xi R) &= 0, \\ GP + FQ + CR + \frac{K}{\theta}(\xi Q - \eta P) &= 0, \end{aligned}$$

or eliminating (P, Q, R) , we have

$$\begin{vmatrix} A & H - \frac{K\zeta}{\theta} & G + \frac{K\eta}{\theta} \\ H + \frac{K\zeta}{\theta} & B & F - \frac{K\xi}{\theta} \\ G - \frac{K\eta}{\theta} & F + \frac{K\xi}{\theta} & C \end{vmatrix} = 0,$$

that is

$$ABC - AF^2 - BG^2 - CH^2 + 2FGH + \frac{K^2}{\theta^2}(A, \dots | \xi, \eta, \zeta)^2 = 0,$$

or, reducing,

$$ABC - AF^2 - BG^2 - CH^2 + 2FGH = K^2,$$

that is

$$\theta^2 + (A, \dots | \xi, \eta, \zeta)^2 = 0,$$

as before.

I reproduce, as follows, a fundamental formula of Aronhold's Memoir. Consider the function

$$\varpi = \frac{1}{\sqrt{-(A, \dots | u, v, w)^2}} \log \frac{(a, \dots | x_1, y_1, z_1 | x, y, z)}{ux + vy + wz},$$

where x_1, y_1, z_1 (corresponding to x, y, z in the former part of this paper) are determined by the conditions

$$(a, \dots | x_1, y_1, z_1)^2 = 0, \quad ux_1 + vy_1 + wz_1 = 0,$$

so that putting

$$\theta = \sqrt{-(A, \dots | u, v, w)^2},$$

and

$$P_1 = ax_1 + hy_1 + gz_1, \quad Q_1 = hx_1 + by_1 + fz_1, \quad R_1 = gx_1 + fy_1 + cz_1,$$

we now have

$$\begin{aligned} -\theta x_1 + R_1 v - Q_1 w &= 0, \\ -\theta y_1 + P_1 w - R_1 u &= 0, \\ -\theta z_1 + Q_1 u - P_1 v &= 0, \end{aligned}$$

and the value of ϖ is

$$\varpi = \frac{1}{\theta} \log \frac{P_1x + Q_1y + R_1z}{ux + vy + wz}.$$

Treating (x, y, z) as independent variables and differentiating, we have

$$\begin{aligned} d\varpi &= \frac{1}{\theta} \left\{ \frac{P_1dx + Q_1dy + R_1dz}{P_1x + Q_1y + R_1z} - \frac{u dx + v dy + w dz}{ux + vy + wz} \right\} \\ &= \frac{1}{\theta} \cdot \frac{(xdy - ydx)(Q_1u - P_1v) + (ydz - zdz)(R_1v - Q_1w) + (zdx - xdz)(P_1w - R_1u)}{(P_1x + Q_1y + R_1z)(ux + vy + wz)} \\ &= \frac{x_1(ydz - zdz) + y_1(zdx - xdz) + z_1(xdy - ydx)}{(P_1x + Q_1y + R_1z)(ux + vy + wz)}, \end{aligned}$$

or, what is the same thing, if

$$P = ax + hy + gz, \quad Q = hx + by + fz, \quad R = gx + fy + cz,$$

so that

$$P_1x + Q_1y + R_1z = (a, \dots | x, y, z | x_1, y_1, z_1) = Px_1 + Qy_1 + Rz_1,$$

then we have

$$d\varpi = \frac{(ydz - zdz)x_1 + (zdx - xdz)y_1 + (xdy - ydx)z_1}{(ux + vy + wz)(Px_1 + Qy_1 + Rz_1)}.$$

Suppose now that (x, y, z) are connected by the equation

$$(a, \dots | x, y, z)^2 = 0,$$

we have

$$Px + Qy + Rz = 0, \quad Pdx + Qdy + Rdz = 0,$$

and thence

$$ydz - zdz = \theta P, \quad zdx - xdz = \theta Q, \quad xdy - ydx = \theta R,$$

and consequently

$$d\varpi = \frac{ydz - zdz}{(ux + vy + wz)P} = \frac{zdx - xdz}{(ux + vy + wz)Q} = \frac{xdy - ydx}{(ux + vy + wz)R},$$

or selecting the value

$$d\varpi = \frac{zdx - xdz}{(ux + vy + wz)Q} = \frac{zdx - xdz}{(ux + vy + wz)(hx + by + fz)},$$

and writing

$$\frac{x}{z} = X, \quad \frac{y}{z} = Y,$$

we have

$$d\varpi = \frac{z^2 dX}{(ux + vy + wz)(hx + by + fz)} = \frac{dX}{(uX + vY + w)(hX + bY + f)},$$

where X and Y are connected by the equation

$$(a, \dots | X, Y, 1)^2 = 0,$$

that is, Y is a given quadric radical function of X . Hence integrating and restoring for ϖ its original value, but writing therein $\frac{x}{z} = X$ and $\frac{y}{z} = Y$, we have

$$\int \frac{dX}{(uX + vY + w)(hX + bY + f)} = \frac{1}{\sqrt{-(A, \dots | u, v, w)^2}} \log \frac{(a, \dots | x_1, y_1, z_1 | X, Y, 1)}{uX + vY + w},$$

where, as just mentioned, Y is a given quadric radical function of X determined by the equation

$$(a, b, c, f, g, h | X, Y, 1)^2 = 0,$$

and the constants x_1, y_1, z_1 are such that

$$(a, \dots | x_1, y_1, z_1)^2 = 0, \quad ux_1 + vy_1 + wz_1 = 0,$$

the ratios of these quantities being therefore determinate; there would, it is clear, be no loss of generality in assuming $z_1 = 1$. This is Aronhold's Theorem I.

2, Stone Buildings, W.C., October 23, 1862.

372.

ON THE RECIPROCATION OF A QUARTIC DEVELOPABLE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vii. (1866), pp. 87–92.]

It is interesting to consider in a particular case the system of equations which shows *à posteriori* that the reciprocal of a torse (developable surface) is a curve (curve in space); and the reciprocal system which shows that the reciprocal of a curve is a torse.

Using (a, b, c, d) and (x, y, z, w) for the reciprocal coordinates, it will be convenient to collect the different equations as follows:

$$a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2 = 0, \quad (1)$$

$$\begin{cases} ad^2 - 3bcd + 2c^3 + \lambda x = 0, \\ -3acd + 6b^2d - 3bc^2 + \lambda y = 0, \\ -3abd + 6ac^2 - 3b^2c - \lambda z = 0, \\ a^2d - 3abc + 2b^3 + \lambda w = 0, \end{cases} \quad (2)$$

$$ax + by + cz + dw = 0, \quad (3)$$

$$3xz - y^2 = 0, \quad yz - 9xw = 0, \quad 3yw - z^2 = 0, \quad (4)$$

$$\begin{cases} 3pz - 9qw + a = 0, \\ -2py + qz + 3rw + b = 0, \\ 3px + qy - 2rz + c = 0, \\ -9qx + 3ry + d = 0. \end{cases} \quad (5)$$

This being so, the equation (1) belongs to a quartic torse, the reciprocal whereof is the skew cubic determined by the equations (4): and we have to show *à posteriori* that this is so.

First if the torse is given, then the reciprocal figure is the envelope of the plane $ax + by + cz + dw = 0$, (3), where (x, y, z, w) are the coordinates and (a, b, c, d) are regarded as variable parameters connected by the equation (1); we thence obtain the equations (2), where λ is an arbitrary multiplier; and from the equations (1), (2), and (3), we have to eliminate a, b, c, d, λ . The equation (3) is at once seen to be included in the equations (1) and (2); and the elimination would give only a single equation between the (x, y, z, w) —since however the equation (1) is that of a torse, we know that the elimination must give two equations, or more accurately a two-fold relation (represented, as in the present case it happens, by three equations) between the coordinates (x, y, z, w) .

Putting for shortness

$$\square = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2,$$

and substituting for $\lambda x, \lambda y, \lambda z, \lambda w$ their values from the equations (1), we have identically

$$\lambda^2(3xz - y^2) = -(bd - c^2)\square,$$

$$\lambda^2(yz - 9xw) = -(ad - bc)\square,$$

$$\lambda^2(3yw - z^2) = -(ac - b^2)\square;$$

and hence (since $\square = 0$) we have between (x, y, z, w) the equations (4), showing that the reciprocal figure is a skew cubic.

Secondly, let the skew cubic be given; then the reciprocal figure is the envelope of the plane $ax + by + cz + dw = 0$, (3), where now (a, b, c, d) are the coordinates and (x, y, z, w) are regarded as variable parameters connected by the equations (4): we thence obtain the equations (5) containing the arbitrary multipliers p, q, r : and from the equations (3), (4), (5) we have to eliminate x, y, z, w, p, q, r . The equation (3) is at once seen to be included in the equations (4) and (5): since however the equations (4) are those of a curve, we know that the elimination must give a single equation between (a, b, c, d) .

The equations (4) are satisfied if we write therein

$$x : y : z : w = 1 : \theta : \theta^2 : \frac{1}{3}\theta^3,$$

and substituting these values of x, y, z, w in the equations (5), these equations give (a, b, c, d) in terms of θ, p, q, r , and we thence find identically

$$\begin{aligned} ac - b^2 &= -\theta^2(p - 2q\theta + r\theta^2)^2, \\ bc - ad &= -2\theta(p - 2q\theta + r\theta^2)^2, \\ bd - c^2 &= -(p - 2q\theta + r\theta^2)^2, \end{aligned}$$

that is

$$ac - b^2 : ad - bc : bd - c^2 = \theta^2 : -2\theta : 1,$$

or we have

$$(ad - bc)^2 - 4(ac - b^2)(bd - c^2) = 0,$$

which is in fact the equation

$$\square = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2 = 0, \quad (1)$$

and thus the reciprocal is in fact the quartic torse, given by the equation (1).

The equations (3), (4), (5) lead to

$$x\delta a + y\delta b + z\delta c + w\delta d = 0;$$

but (a, b, c, d) are in consequence of these equations connected by a single equation, viz. the equation (1); the equation just obtained is thus the only relation between the differentials $(\delta a, \delta b, \delta c, \delta d)$: it is clear from the equations (2) that this is in fact the equation

$$\delta\square = \delta_a\square\delta a + \delta_b\square\delta b + \delta_c\square\delta c + \delta_d\square\delta d = 0.$$

The equations (1), (2), (3) lead in like manner to

$$a\delta x + b\delta y + c\delta z + d\delta w = 0,$$

which in virtue of the equations (5) is equivalent to

$$p\delta(3xz - y^2) + q\delta(yz - 9xw) + r\delta(3yw - z^2) = 0,$$

viz. it is a consequence of

$$\delta(3xz - y^2) = 0, \quad \delta(yz - 9xw) = 0, \quad \delta(3yw - z^2) = 0.$$

But inasmuch as (x, y, z, w) are connected by a two-fold relation, the equations (1), (2), (3) must lead to one other linear relation between the differentials $(\delta x, \delta y, \delta z, \delta w)$: and I proceed to show that this is so.

Differentiating we have

$$\begin{aligned} d^2\delta a - 3cd\delta b + (-3bd + 6c^2)\delta c + (2ad - 3bc)\delta d + \lambda dx + x d\lambda &= 0, \\ -3cd\delta a + (12bd - 3c^2)\delta b + (-3ad - 6bc)\delta c + (-3ac + 6b^2)\delta d + \lambda dy + y d\lambda &= 0, \\ (-3bd + 6c^2)\delta a + (-3ad - 6bc)\delta b + (12ac - 3b^2)\delta c - 3ab\delta d + \lambda dz + z d\lambda &= 0, \\ (2ad - 3bc)\delta a + (-3ac + 6b^2)\delta b - 3ab\delta c + a^2\delta d + \lambda dw + w d\lambda &= 0. \end{aligned}$$

Consider the matrix

$$\begin{pmatrix} d^2 & -3cd & -3bd + 6c^2 & 2ad - 3bc \\ -3cd & 12bd - 3c^2 & -3ad - 6bc & -3ac + 6b^2 \\ -3bd + 6c^2 & -3ad - 6bc & 12ac - 3b^2 & -3ab \\ 2ad - 3bc & -3ac + 6b^2 & -3ab & a^2 \end{pmatrix}$$

and suppose for a moment that the determinant formed therewith is $= K$; suppose also that the reciprocal matrix is

$$\begin{pmatrix} \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \mathfrak{L} \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \mathfrak{M} \\ \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \mathfrak{N} \\ \mathfrak{L} & \mathfrak{M} & \mathfrak{N} & \mathfrak{D} \end{pmatrix}.$$

Then we have

$$K \delta a + d\lambda(\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z + \mathfrak{L}w) + \lambda(\mathfrak{A}dx + \mathfrak{H}dy + \mathfrak{G}dz + \mathfrak{L}dw) = 0,$$

with the similar equations involving b, c, d and $(\mathfrak{H}, \mathfrak{B}, \mathfrak{F}, \mathfrak{M})$, $(\mathfrak{G}, \mathfrak{F}, \mathfrak{C}, \mathfrak{N})$, $(\mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{D})$ respectively.

But, substituting for (x, y, z, w) their values from the equations (2), it is easy to see that we have

$$\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z + \mathfrak{L}w = -\frac{1}{3\lambda} Ka,$$

the last-mentioned equation thus is

$$K(\delta a - \frac{1}{3} a \frac{d\lambda}{\lambda}) + \lambda(\mathfrak{A} \delta x + \mathfrak{H} \delta y + \mathfrak{G} \delta z + \mathfrak{L} \delta w) = 0.$$

But K is $= 27\Box$ (see my paper “On Certain Developable Surfaces,” *Quarterly Mathematical Journal*, t. vi. 1864, pp. 108–126, [344]), which is $= 0$, and we thus have

$$\begin{aligned} \mathfrak{A} \delta x + \mathfrak{H} \delta y + \mathfrak{G} \delta z + \mathfrak{L} \delta w &= 0, \\ \mathfrak{H} \delta x + \mathfrak{B} \delta y + \mathfrak{F} \delta z + \mathfrak{M} \delta w &= 0, \\ \mathfrak{G} \delta x + \mathfrak{F} \delta y + \mathfrak{C} \delta z + \mathfrak{N} \delta w &= 0, \\ \mathfrak{L} \delta x + \mathfrak{M} \delta y + \mathfrak{N} \delta z + \mathfrak{D} \delta w &= 0. \end{aligned}$$

But observing that in virtue of the equation $K = 0$, we have

$$\mathfrak{A} : \mathfrak{H} : \mathfrak{G} : \mathfrak{L} = \mathfrak{H} : \mathfrak{B} : \mathfrak{F} : \mathfrak{M} = \mathfrak{G} : \mathfrak{F} : \mathfrak{C} : \mathfrak{N} = \mathfrak{L} : \mathfrak{M} : \mathfrak{N} : \mathfrak{D},$$

these are, as they should be, one and the same equation.

The values of the coefficients $\mathfrak{A}, \mathfrak{B}$, etc. are given, p. 112 of the paper just referred to, viz. writing $\Box$ in place of U , we have $\mathfrak{A} = 3X^2 - 4a^2\Box$, etc. where

$$\begin{aligned} X &= a^2d - 3abc + 2b^3, \\ Y &= abd - 2ac^2 + b^2c, \\ Z &= -acd + 2b^2d - bc^2, \\ W &= -ad^2 + 3bcd - 2c^3, \end{aligned}$$

and writing $\Box = 0$, the values each divided by 3, are simply

$$\begin{array}{cccc} X^2 & XY & XZ & XW \\ YX & Y^2 & YZ & YW \\ ZX & ZY & Z^2 & ZW \\ WX & WY & WZ & W^2 \end{array}$$

so that each of the four equations in fact becomes

$$X \delta x + Y \delta y + Z \delta z + W \delta w = 0,$$

or multiplying by 3, and attending to the equations (2), this is

$$-3w \delta x + z \delta y - y \delta z + 3x \delta w = 0.$$

This should be a consequence of the equations

$$\delta(3xz - y^2) = 0, \quad \delta(yz - 9xw) = 0, \quad \delta(3yw - z^2) = 0,$$

that is, we should be able from the first three to deduce the fourth equation in the system

$$\begin{aligned} 3z \delta x - 2y \delta y + 3x \delta z &= 0, \\ -9w \delta x + z \delta y + y \delta z - 9x \delta w &= 0, \\ 3w \delta y - 2z \delta z + 3y \delta w &= 0, \\ -3w \delta x + z \delta y - y \delta z + 3x \delta w &= 0, \end{aligned}$$

or we ought to have

$$\begin{vmatrix} 3z & -2y & 3x & 0 \\ -9w & z & y & -9x \\ 0 & 3w & -2z & 3y \\ -3w & z & -y & 3x \end{vmatrix} = 0;$$

but expanding, this is

$$-6(81x^2w^2 - 54xyzw + 12xz^3 + 12y^3w - 3y^2z^2) = 0,$$

or

$$(yz - 9xw)^2 - 4(xz - y^2)(yw - z^2) = 0,$$

which is true in virtue of the relations (4). Or what is the same thing, we may show without difficulty that the equation

$$-3w \delta x + z \delta y - y \delta z + 3x \delta w = 0,$$

is satisfied by writing therein $x : y : z : w = 1 : \theta : \theta^2 : \frac{1}{3}\theta^3$.

I remark that in general, if $\square = \phi(a, b, c, d) = 0$ is the equation of a torse, then for finding the reciprocal curve, we have

$$\begin{aligned} \square &= 0, \quad ax + by + cz + dw = 0, \\ \delta_a \square + \lambda x &= 0, \\ \delta_b \square + \lambda y &= 0, \\ \delta_c \square + \lambda z &= 0, \\ \delta_d \square + \lambda w &= 0, \end{aligned}$$

and that from these equations we deduce not only

$$a \delta x + b \delta y + c \delta z + d \delta w = 0,$$

but also the equation

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N} \mid \alpha, \beta, \gamma, \delta \mid \delta x, \delta y, \delta z, \delta w) = 0,$$

where $\mathfrak{A}, \dots$ are the inverse system derived from the second differential coefficients of $\square$: $(\alpha, \beta, \gamma, \delta)$ are arbitrary coefficients, introduced only for symmetry, and there is no real loss of generality in reducing all but one of them to zero, and so reducing the equation for example to the form

$$\mathfrak{A} \delta x + \mathfrak{H} \delta y + \mathfrak{G} \delta z + \mathfrak{L} \delta w = 0.$$

The existence of the two linear equations between $(\delta x, \delta y, \delta z, \delta w)$ proves that (x, y, z, w) are connected by a two-fold relation, that is, that the reciprocal of the given torse is a curve.

Cambridge, January 26, 1865.

373.

ON A SPECIAL SEXTIC DEVELOPABLE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vii. (1866), pp. 105–113.]

The present paper contains some investigations in relation to the special sextic developable or torse

$$(ae - 4bd)^3 - 27(-ad^2 - b^2e)^2 = 0,$$

considered Nos. 26 to 35 of my paper “On Certain Developable Surfaces,” *Quarterly Mathematical Journal*, t. vi. (1864), pp. 108–126, [344].

The cuspidal curve is

$$ae - 4bd = 0, \quad ad^2 + b^2e = 0,$$

and the nodal curve is

$$ae + 2bd = 0, \quad ad^2 - b^2e = 0,$$

viz. to put this in evidence, the equation is to be written in the form

$$(ae + 2bd)^2(ae - 16bd) - 27(ad^2 - b^2e)^2 = 0.$$

The coordinates of a point on the cuspidal curve may be taken to be

$$a = 2, \quad b = -t, \quad d = +t^3, \quad e = -2t^4,$$

and then if A, B, D, E are current coordinates, and $\alpha, \beta, \delta, \epsilon$ arbitrary coefficients, the equation of a plane through the tangent line is

$$\begin{vmatrix} A & B & D & E \\ 2 & -t & +t^3 & -2t^4 \\ \cdot & -1 & +3t^2 & -8t^3 \\ \alpha & \beta & \delta & \epsilon \end{vmatrix} = 0$$

which is

$$\begin{aligned} &+ (A\beta - B\alpha)(-t^4) \\ &+ (A\delta - D\alpha)(-t^2) \\ &+ (AE - E\alpha)(-t^6) \\ &+ (B\delta - D\beta)(-5t^3) \\ &+ (D\epsilon - E\delta)(-1) = 0, \end{aligned}$$

or, what is the same thing,

$$\alpha(Bt^4 + 3Dt^2 + Et^6) + \beta(-At^4 + 8Dt^3 + 3Et^5) + \delta(-3At^2 - 8Bt^3 + E) + \epsilon(-At^6 - 3Bt^5 - D) = 0,$$

and by equating to zero the coefficients $\alpha, \beta, \delta, \epsilon$, we have four equations which are easily seen to reduce themselves to two equations only, and which are in fact the equations of the tangent line, the equations of this line may therefore be taken to be

$$At^2 + 3Bt + D = 0, \quad 3At^3 + 8Bt^2 - E = 0.$$

The coordinates of a point in the nodal curve may be taken to be

$$a = \sqrt{2}, \quad b = \tau, \quad d = -\tau^3, \quad e = \sqrt{2}\tau^4,$$

and substituting these values in the place of A, B, D, E in the equations of the tangent line, we have

$$\sqrt{2}t^2 + 3t\tau - \tau^3 = 0,$$

$$3\sqrt{2}t^3 + 8t^2\tau - \sqrt{2}\tau^4 = 0,$$

or, what is the same thing,

$$\begin{aligned}\tau^3 - 3t\tau + \sqrt{2}t^2 &= 0, \quad \text{i.e. } (\tau + \sqrt{2}t)(\tau^2 - \sqrt{2}t\tau - t^2) = 0, \\ \tau^4 - 4\sqrt{2}t^3\tau - 3t^4 &= 0, \quad (\tau^2 + \sqrt{2}t\tau - t^2)(\tau^2 - \sqrt{2}t\tau - t^2) = 0,\end{aligned}$$

so that the equations are satisfied by the values of τ given by the equation

$$\tau^2 - \sqrt{2}t\tau - t^2 = 0,$$

that is, by the values

$$\tau = \frac{1 \pm \sqrt{3}}{\sqrt{2}} t,$$

which belong to the points where the tangent line meets the nodal curve. Call these values τ_1 and τ_2 ; then considering a, b as current coordinates, the values of $a : b$ belonging to the point where the tangent line meets the cuspidal curve considered as three coincident points, and to the points where it meets the nodal curve, are given by the equation

$$(2b + at)^3(b\sqrt{2} - a\tau_1)(b\sqrt{2} - a\tau_2) = 0,$$

that is

$$(2b + at)^3(2b^2 - 2abt - a^2t^2) = 0,$$

or say

$$(at + 2b)^3(a^2t^2 + 2abt - 2b^2) = 0.$$

I proceed to find the intersections of the tangent with the Prohessian: for this purpose putting for a moment in the last-mentioned equation x for at and y for b , this is

$$(x + 2y)^3(x^2 + 2xy - 2y^2) = 0,$$

or, if in the place of $(x + y)$ we write x , this is

$$(x + y)^3(x^2 - 3y^2) = 0,$$

and the Hessian of this is easily found to be

$$(x + y)^3(3x^2 + 8xy + 4y^2),$$

whence, replacing x by $(x + y)$, the Hessian of

$$(x + 2y)^3(x^2 + 2xy - 2y^2)$$

is

$$(x + 2y)^3(3x^2 + 14xy + 18y^2).$$

We have thus

$$3x^2 + 14xy + 18y^2 = 0;$$

that is

$$3x + (7 \pm \sqrt{-5})y = 0,$$

or

$$3at + (7 \pm \sqrt{-5})b = 0,$$

and therefore

$$\frac{b}{a} = \frac{-3t}{7 \pm \sqrt{-5}} = \frac{-3(7 \mp \sqrt{-5})}{54} t = -\frac{7 \mp \sqrt{-5}}{18} t,$$

or putting

$$n_1 = -\frac{7 + \sqrt{-5}}{18}, \quad n_2 = -\frac{7 - \sqrt{-5}}{18},$$

so that $n_1 + n_2 = -\frac{7}{9}$, $n_1 n_2 = \frac{1}{6}$, and n_1, n_2 are the roots of the equation $18n^2 + 14n + 3 = 0$, then we have $\frac{b}{a} = n_1 t$ or $n_2 t$, say $\frac{b}{a} = n_i t$, or assuming $a = 1$, then $b = n_i t$.

But the equations of the tangent line being

$$at^2 + 3bt + d = 0, \quad 3at^3 + 8bt^2 - e = 0,$$

we have thus

$$\begin{aligned} a &= 1, & a &= 1, \\ b &= n_1 t, & b &= n_2 t, \\ d &= (-1 - 3n_1) t^2, & d &= (-1 - 3n_2) t^2, \\ e &= (3 + 8n_1) t^3, & e &= (3 + 8n_2) t^3, \end{aligned}$$

as the coordinates of the required points, viz. the tangent line meets the Prohessian, in the point on the cuspidal edge considered as 6 points, in two points on the nodal curve and in the last-mentioned 2 points; $6 + 2 + 2 = 10$ the order of the Prohessian.

The foregoing equations give

$$ae - 6bd = 18n_i^2 + 14n_i + 3 = 0, \quad (\text{in virtue of } 18n_i^2 + 14n_i + 3 = 0),$$

so that the two points in question are the intersections of the tangent line with the surface $ae - 6bd = 0$.

If we consider the intersection of this surface with the torse

$$(ae - 4bd)^3 - 27(-ad^2 - b^2e)^2 = 0,$$

the equation $ae - 6bd = 0$ gives

$$(ae - 4bd)^3 = (2bd)^3 = 8b^3d^3, \quad bd = \frac{1}{6}aeb^2d^2,$$

and thence

$$(ae - 4bd)^3 = 81(ad^2 + b^2e)^2 = 0,$$

that is

$$\begin{aligned} 4a^2b^2d^4e - 81(ad^2 + b^2e)^2 &= 0; \\ 81a^2d^4 - 158ab^2d^2e + 81b^4e^2 &= 0, \end{aligned}$$

an equation which should agree with the above.

In fact writing

$$x = \frac{1}{81}(144n_i - 23), \quad x = \frac{1}{81}(144n_i - 23),$$

the equation $18n_i^2 + 14n_i + 3 = 0$ is $(18n_i + 7)^2 + 5 = 0$; that is, $(144n_i + 56)^2 + 320 = 0$, but $144n_i + 56 = 81x + 79$, or the equation becomes $(81x + 79)^2 + 320 = 0$, that is

$$81^2x^2 + 81 \cdot 158x + 81^2 = 0, \quad \text{or} \quad 81x^2 + 158x + 81 = 0,$$

which is right.

Consider in like manner the intersection of the torse with the surface $ae - \lambda bd = 0$, where λ is a given constant coefficient; we have

$$(ae - 4bd)^3 = (\lambda - 4)^3 b^3 d^3 = \frac{(\lambda - 4)^3}{\lambda} aeb^2 d^2,$$

and therefore

$$(\lambda - 4)^3 aeb^2 d^2 - 27\lambda(ad^2 + b^2e)^2 = 0,$$

that is

$$27\lambda a^2 d^4 + [54\lambda - (\lambda - 4)^3] ab^2 d^2 e + 27\lambda b^4 e^2 = 0,$$

which gives

$$ad^2 - \theta_1 b^2 e = 0, \quad \text{or} \quad ad^2 - \theta_2 b^2 e = 0,$$

if θ_1, θ_2 are the roots of

$$27\lambda\theta^2 + [54\lambda - (\lambda - 4)^3]\theta + 27\lambda = 0.$$

The surfaces $ae - \lambda bd = 0$, $ad^2 - \theta_1 b^2 e = 0$ have in common the two lines $(a = 0, b = 0)$ and $(d = 0, e = 0)$, and they intersect besides in a quartic curve. And so for the surfaces $ae - \lambda bd = 0$, $ad^2 - \theta_2 b^2 e = 0$. That is, the surface $ae - \lambda bd = 0$ intersects the torse $(ae - 4bd)^3 - 27(-ad^2 - b^2 e)^2 = 0$, in the line $a = 0, b = 0$ twice, in the line $d = 0, e = 0$ twice, and in two quartic (excubo-quartic) curves. The two quartic curves become identical, if

$$(54\lambda)^2 = [54\lambda - (\lambda - 4)^3]^2,$$

and therefore, if either

$$\pm 54\lambda = 54\lambda - (\lambda - 4)^3,$$

that is

$$(\lambda - 4)^3 = 0,$$

which gives the cuspidal curve; or else if

$$(\lambda - 4)^3 - 108\lambda = 0,$$

that is

$$\lambda^3 - 12\lambda^2 - 60\lambda - 64 = (\lambda + 2)^2(\lambda - 16) = 0.$$

$(\lambda + 2)^2 = 0$ or $\lambda = -2$ gives the nodal curve: $\lambda - 16 = 0$ gives $ae - 16bd = 0$, a surface which intersects the developable in the line $a = 0, b = 0$ twice, in the line $d = 0, e = 0$ twice, and in the two coincident quartic (excubo-quartic) curves given by the equations $ae - 16bd = 0$, $ad^2 - b^2 e = 0$. As a verification, I remark, that the surface $ad^2 - b^2 e = 0$ combined with the developable gives

$$(ae - 4bd)^3 - 27(ad^2 + b^2 e)^2 = (ae - 4bd)^3 - 108ab^2 d^2 e = 0,$$

that is $(ae + 2bd)^2(ae - 16bd) = 0$, or it meets the developable in its curve of intersection with $ae + 2bd = 0$ twice, and in its curve of intersection with $ae - 16bd = 0$; that is, in the line $a = 0, b = 0$ three times, in the line $d = 0, e = 0$ three times, in the nodal quartic $ae + 2bd = 0$, $ad^2 - b^2 e = 0$ twice, and in the quartic $ae - 16bd = 0$, $ad^2 - b^2 e = 0$ once; $3 + 3 + 8 + 4 = 18$, the order of the complete intersection.

Greenwich, January 4, 1864.

In my theory of the singularities of curves and torsos, *Liouville*, t. x. (1845) pp. 245–250, [30], translated under the title “On Curves of Double Curvature and Developable Surfaces,” *Cambridge and Dublin Mathematical Journal*, t. vi. (1850), pp. 18–22, [83], I omitted to take account of a noteworthy singularity, viz. this is the stationary tangent line; or when the system has three consecutive points in a line, or, what is the same thing, three consecutive planes through a line. I reproduce the theory with this addition as follows. We have

m , the order of the system, = order of the curve,

r , „ rank of the system, = class of curve, = order of torse,

n , „ class of the system, = class of torse.

α , „ number of stationary planes,

β , „ „ stationary points,

$\mathfrak{H}$, „ „ stationary lines,

g , „ „ lines in two planes,

h , „ „ lines through two points,

x , „ „ points in two lines,

y , „ „ planes through two lines.

This being so, the section of the torse by an arbitrary plane is a plane curve for which

r is the order,

n „ class,

x „ number of nodes,

$m + \mathfrak{H}$ „ „ cusps,
 g „ „ double tangents,
 α „ „ inflexions;

and we have thence Plücker's six equations, which may be considered as included in the three equations

$$\begin{aligned} n &= r(r-1) - 2x - 3(m + \mathfrak{H}), \\ \alpha &= 3r(r-2) - 6x - 8(m + \mathfrak{H}), \\ r &= n(n-1) - 2g - 3\alpha. \end{aligned}$$

Similarly considering the cone standing on the curve and having an arbitrary point for vertex, then for this cone

m is the order,
 r „ class,
 h „ number of nodal lines,
 β „ „ cuspidal lines,
 y „ „ double tangent planes,
 $n + \mathfrak{H}$ „ „ inflexions;

and we have Plücker's six equations, which may be considered as included in the three equations

$$\begin{aligned} m &= r(r-1) - 2y - 3(n + \mathfrak{H}), \\ \beta &= 3r(r-2) - 6y - 8(n + \mathfrak{H}), \\ r &= m(m-1) - 2h - 3\beta. \end{aligned}$$

These two systems constitute together a system of six equations between the ten quantities $m, r, n, \alpha, \beta, \mathfrak{H}, g, h, x, y$. Considering $m, r, x, \mathfrak{H}$ as arbitrary, the six equations determine the remaining quantities n, α, β, h, y .

The curve

$$ae - 4bd + 3c^2 = 0, \quad ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

is a sextic curve, the edge of regression of the sextic torse

$$(ae - 4bd + 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

and we have in this case, as is well known,

$$m, r, n, \alpha, \beta, \mathfrak{H}, g, h, x, y = 6, 6, 4, 0, 4, 0, 3, 6, 4, 6.$$

But putting as above $c = 0$, then instead of the sextic curve we have the excubo-quartic curve $ae - 4bd = 0$, $ad^2 + b^2e = 0$, which is a curve having two stationary tangents, viz. these are the lines ($a = 0, b = 0$) and ($d = 0, e = 0$), which are in fact given along with the curve, by the foregoing equations $ae - 4bd = 0$, $ad^2 + b^2e = 0$. We have in this case $\mathfrak{H} = 2$, and the system is thus found to be

$$m, r, n, \alpha, \beta, \mathfrak{H}, g, h, x, y = 4, 6, 4, 0, 0, 2, 3, 3, 4, 4,$$

it was in fact the consideration of this case which led me to take account of the new singularity of the stationary tangent lines.

I take the opportunity of referring to a most valuable and interesting paper by Schwarz, “De superficiebus in planum explicabilibus primorum septem ordinum,” *Crelle*, t. lxiv. (1864), pp. 1–16. The author, after referring to my paper “On the developable derived from an equation of the fifth order,” *Cambridge and Dublin Mathematical Journal*, t. v. (1850), pp. 152–159, [86], enters into the enquiry there suggested as to the means of ascertaining the degree of the ‘planarity’ of a developable surface. He starts from certain theorems derived from Riemann's theory of transcendental functions, viz.: If an algebraical (plane) curve

of the order r has $\frac{1}{2}(r-1)(r-2) - p$ double points (nodes or cusps), then the coordinates of a point of the curve may be expressed rationally:

If $p = 0$, that is, if the curve has the maximum number of double points, by a single parameter.

If $p = 1$, by a single parameter, and the square root of a cubic or quartic function of this parameter.

If $p = 2$, by a single parameter, and the square root of a quintic or sextic function of this parameter.

If $p > 2$, by a parameter ξ , and an algebraical function thereof η ; where ξ, η are connected by an equation of the order $\frac{1}{2}(p+3)$ or $\frac{1}{2}(p+2)$ according as p is odd or even.

These principles establish a division of plane curves into algebraical classes: all plane curves (other than the generating lines) situate on a ruled surface, belong to the same algebraical class, and the surface itself belongs to the same class. Hence, if on a ruled surface there is either a right line which is not a generating line (this cannot be the case for developables) or a conic, or a cubic having a double point, or any other plane curve having the maximum number of double points, the surface belongs to the class for which $p = 0$; and in the case of a developable surface the equation of the tangent plane may be rationally expressed by means of a single parameter; that is, the degree of the planarity is $= 1$, or the surface is planar. This leads to the conclusion, that the developable surfaces or torses of the orders 4, 5, 6 and 7 are all of them planar.

The author points out that the ‘special quintic developable’ of my paper first above referred, (viz. that obtained by writing $b = 0$ in the equation of the sextic developable) is in fact the general developable of the fifth order, or quintic torse.

The foregoing theorem, that for a curve which has the maximum number of double points, the coordinates may be expressed rationally by a single parameter, admits of a very simple algebraical proof, as is shown in the paper by Clebsch “Ueber Curven deren coordinaten rationale Functionen eines Parameters sind,” *Crelle*, t. lxiv. (1864), pp. 43–65. In another paper by the same author, “Ueber die Singularitäten algebraischer Curven,” pp. 98–100, it is remarked that if in any plane curve we have m the order, n the class, δ the number of nodes, κ of cusps, τ of double tangents, ι of inflexions, then as a deduction from Riemann’s principles, but also at once obtainable from Plücker’s equations, we have

$$\frac{1}{2}(m-1)(m-2) - \delta - \kappa = \frac{1}{2}(n-1)(n-2) - \tau - \iota;$$

and moreover if from a given curve we derive in any manner another curve, such that to each tangent (or point) of the first curve there corresponds a single point (or tangent) of the second curve, then in the second curve the expression

$$\frac{1}{2}(m'-1)(m'-2) - \delta' - \kappa' = \frac{1}{2}(n'-1)(n'-2) - \tau' - \iota',$$

has the same value as in the first curve.

The like property exists for curves in space—viz. taking account as above of the new singularity of the stationary lines, then we have

$$\begin{aligned} & \frac{1}{2}(m-1)(m-2) - h - \beta \\ &= \frac{1}{2}(r-1)(r-2) - y - n - \mathfrak{H} \\ &= \frac{1}{2}(r-1)(r-2) - x - m - \mathfrak{H} \\ &= \frac{1}{2}(n-1)(n-2) - g - \alpha, \end{aligned}$$

which equations are in fact at once deducible from the above-mentioned system of six equations between the quantities $m, r, n, \alpha, \beta, \mathfrak{H}, g, h, x, y$, and may if we please be taken for equations of the system.

If from a given curve and torse we derive a second curve and torse, in such manner that to each point (or plane) of the first figure there corresponds a single plane (or point) of the second figure—then the corresponding expressions $\frac{1}{2}(m'-1)(m'-2) - h' - \beta'$, etc., have the same value for the second as for the first figure.

Cambridge, April 11, 1865.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. vii. (1866), pp. 212–223.]

The theory of the singularities of a plane curve was first established by Plücker in his great work the *Theorie der Algebraischen Curven*, (1839), where he establishes, in regard to the ordinary singularities, a system of six equations; viz. if we have

m , the order of the curve,
 n , „ class,
 δ , „ number of double points,
 κ , „ „ cusps,
 τ , „ „ double tangents,
 ι , „ „ inflexions,

then Plücker's six equations are

$$\begin{aligned} n &= m(m-1) - 2\delta - 3\kappa, \\ \iota &= 3m(m-2) - 6\delta - 8\kappa, \\ \tau &= \frac{1}{2}m(m-2)(m^2-9) - (m^2-m-6)(2\delta+3\kappa) + 2\delta(\delta-1) + 6\delta\kappa + \frac{9}{2}\kappa(\kappa-1), \\ m &= n(n-1) - 2\tau - 3\iota, \\ \kappa &= 3n(n-2) - 6\tau - 8\iota, \\ \delta &= \frac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2\tau+3\iota) + 2\tau(\tau-1) + 6\tau\iota + \frac{9}{2}\iota(\iota-1), \end{aligned}$$

equivalent to three equations; thus m and (within proper limits) δ and κ may be considered arbitrary, and the first three equations then give n, ι, τ ; and in like manner n and (within proper limits) τ and ι may be considered as arbitrary, and the equations then give m, κ, δ .

I have used the ordinary expressions double points, cusps, double tangents, inflexions; but (using as I have elsewhere done *ineunt* as the correlative of tangent) it would be more precise and symmetrical to say, double ineunts, stationary ineunts, double tangents, and stationary tangents. The double ineunt is called also a node, viz. it is a *crunode*, or an *acnode*, according as the tangents are real or imaginary; and the stationary ineunt, or cusp, considered as (what in the theory of point-coordinates it in fact is) a particular case of the double ineunt, is a *spinode*; to render this notation symmetrical, we require certain new terms, say link, as the correlative to node, and flex as the correlative to cusp; then the double tangent is a link, viz. it is a *colink*, or an *allink* according as the ineunts upon it (points of contact) are real or imaginary; and the stationary tangent (inflexion) or flex, considered as (what in the theory of line-coordinates it in fact is) a particular case of the double tangent, is a *relink*. The ordinary singularities of a plane curve would thus be the node, the cusp, the link, and the flex; but I shall retain the above-mentioned more usual expressions.

Deducible from the six equations, we have

$$\begin{aligned} n - m &= \frac{1}{2}(\iota - \kappa), \\ (n - m)(n + m - 9) &= 2(\tau - \delta), \end{aligned}$$

which are noticed by Plücker; and also the equation

$$\frac{1}{2}(m-1)(m-2) - \delta - \kappa = \frac{1}{2}(n-1)(n-2) - \tau - \iota,$$

recently noticed by M. Clebsch, in connection with Riemann's investigations on the Abelian Integrals; a curve of the order m may have $\frac{1}{2}(m-1)(m-2)$ double points, reckoning the cusp as a double point, and so a curve of the class n may have $\frac{1}{2}(n-1)(n-2)$ double tangents, reckoning the inflexion as a double tangent; the two sides of this equation exhibit therefore, the right-hand side the deficiency of the actual from the possible number of double tangents, and the left-hand side the deficiency of the actual from the possible number of double points; and these two numbers are equal. We have a division into families based on the value of the expressions in question, or say on that of $\frac{1}{2}(m-1)(m-2) - \delta - \kappa$; when this is $= 0$, that is, when the curve has its maximum number of double points (reckoning the cusp as a double point), the coordinates x, y are expressible rationally in terms of a parameter θ ; when the number is $= 1$, they can

be expressed rationally in terms of θ and of the square root of a cubic or a quartic function of θ , etc. etc. It thus appears that as well the number $\delta + \kappa$, as the combinations $2\delta + 3\kappa$ and $6\delta + 8\kappa$ which enter into Plücker's equations, plays an important part in the theory of the curve; the bearing of this remark will be seen in the sequel.

Plücker considers also some of the higher singularities; it will be convenient to mention two of his results.

No. 76, p. 216. If two branches of a curve touch each other, or more generally have a g -pointic intersection, the point in question is equivalent to g double points, and the tangent at this point to g double tangents; hence, if there is no other point singularity, the equations give

$$\begin{aligned} n &= m(m-1) - 2g, \\ \iota &= 3m(m-2) - 6g, \\ \delta + g &= \frac{1}{2}m(m-2)(m^2-9) - (m^2-m-6) \cdot 2g + 2g(g-1), \end{aligned}$$

the last of which may also be written

$$\delta = \frac{1}{2}m(m-2)(m^2-9) - (m^2-m-g-4) \cdot 2g.$$

And Nos. 77–82, pp. 217–222. For a cusp of the second kind, we have

$$\begin{aligned} n &= m(m-1) - 5, \\ \iota &= 3m(m-2) - 15, \\ \delta &= \frac{1}{2}m(m-2)(m^2-9) - (m^2-m-7) \cdot 5; \end{aligned}$$

these equations Plücker establishes by an independent algebraical investigation, and having done so, he remarks that they are deducible from the foregoing ones by writing therein $g = 2\frac{1}{2}$; that is, that the cusp of the second kind $\mathfrak{B}$ may be considered as equivalent to $2\frac{1}{2}$ double points, and the tangent at the cusp to $2\frac{1}{2}$ double tangents. And he thence passes to the cusp of a higher cusp equivalent to $\mu + \frac{1}{2}$ double points and $\mu + \frac{1}{2}$ double tangents. The results in this general case (although not, as in the original case, $g = 2\frac{1}{2}$, established independently) is perfectly correct; but the theory is open to a grave objection.

I remark, that assuming a certain singularity to be equivalent to the numbers δ' of double points, κ' of cusps, τ' of double tangents, and ι' of inflexions, we have in the first instance to determine δ', κ', τ' and ι' in such manner as to give in the class n , and in the numbers ι of inflexions and τ of double tangents, the reductions actually given by the singularity in question. Thus in the case of the cusp of the second kind, we ought to have

$$\begin{aligned} 2\delta' + 3\kappa' &= 5, \\ 6\delta' + 8\kappa' + \iota' &= 15, \\ (m^2 - m - 6)(2\delta' + 3\kappa') - 2\delta'(\delta' - 1) - 6\delta'\kappa' - \frac{9}{2}\kappa'(\kappa' - 1) + \tau' &= (m^2 - m - 7) \cdot 5, \end{aligned}$$

or, what is the same thing,

$$2\delta'(\delta' - 1) + 6\delta'\kappa' + \frac{9}{2}\kappa'(\kappa' - 1) - \tau' = 5;$$

and so in general there are, for the determination of the four quantities $\delta', \kappa', \tau', \iota'$, three equations. In the particular case these are satisfied by the values $\delta' = 2\frac{1}{2}, \kappa' = 0, \tau' = 2\frac{1}{2}, \iota' = 0$, which are Plücker's values; they are also satisfied by the values $\delta' = 1, \kappa' = 1, \tau' = 1, \iota' = 1$, which have the advantage of being integer instead of fractional.

But there is really a further condition to be satisfied, viz. the number $\delta' + \kappa'$ must have a certain definite value dependent on the nature of the singularity; for the case in hand, this is $\delta' + \kappa' = 2$ (I obtain this by the consideration of a quartic curve, having a cusp of the second kind, and also a double point; $\delta + \kappa$ has here its maximum value = 3; and as the double point gives $\delta = 1$, the cusp of the second kind gives $\delta' + \kappa' = 2$); and joining to the former conditions this new condition, we have definitely $\delta' = 1, \kappa' = 1, \tau' = 1, \iota' = 1$. I have elsewhere noticed, [343], that the cusp of the second kind was equivalent to a double point and cusp, and accordingly proposed to call it the node-cusp; but I had not then remarked that it was also necessary to treat the tangent as equivalent to a double tangent and a stationary tangent (or inflexion).

It appears from the foregoing considerations, that any singularity whatever is to be regarded as equivalent, and that in a perfectly definite manner, to a certain number δ' of double points, κ' of cusps, τ' of double tangents, and ι' of inflexions; we have only to ascertain how for any given singularity the values of these numbers are to be ascertained; and when this is done, Plücker's equations will be applicable to any singularities whatever of a plane curve.

At any point of a plane curve there is either one branch, or any number of branches, touching or not touching each other: taking the given point as origin, then for each branch the equation of the curve gives for the ordinate y an expression of the form

$$y = Ax^p + Bx^q + \dots,$$

where the series is arranged in ascending powers of x , and where the coefficients $A, B, \dots$ have definite unique values; and, conversely, that which is given by such expression of y is a branch of the curve. It is assumed that the axis of y , or line $x = 0$, is not a tangent to the curve; this implies that the exponents $p, q, \dots$ are none of them inferior to 1, or, what is the same thing, that the lowest exponent p is $= 1$ at least: it is for the most part convenient to take the axis of x , or line $y = 0$, a tangent to the branch; the lowest exponent p is then > 1 .

The exponents may be all integer, and the branch is then said to be linear; or else the exponents or some of them may be fractional, and the branch is then *superlinear*; viz. in the latter case, assuming that the fractional exponents are all of them in their least terms, and that α is the least common multiple of the denominators (so that the expression for y is a rational function of $x^{1/\alpha}$), then the branch is quadric, cubic, etc. according as we have $\alpha = 2, \alpha = 3$, etc. It is clear that the expression for y has precisely α values, viz. the values obtained by attributing to the radical $x^{1/\alpha}$ each of its α values. Corresponding to each of these α values, we have what I term a partial branch of the curve, so that the quadric branch is made up of two partial branches, the cubic branch of three partial branches and so on; for a linear branch or when $\alpha = 1$, a partial branch is nothing else than the branch itself; and the expression a partial branch will accordingly include the case of a linear branch.

Suppose that at any point of the curve we have two partial branches, belonging or not belonging to the same branch; let these be referred to the same axes, the axis of y not being a tangent to either branch, so that the exponents are none of them < 1 . If in the series for $y_1 - y_2$, (the difference of the two ordinates) the least exponent is $= P$, then (whether P is integer or fractional) the two partial branches are said to have, at the given point, P common points, or, more briefly, to intersect in P points. We may from this definition calculate the number of intersections of two branches with each other, or of a branch with itself; for instance, suppose that at any point of the curve we have ($\alpha = 6$) the sextic branch

$$y = x^{4/6} + x^{7/6} + \dots,$$

we have the six partial branches

$$\begin{aligned} y_1 &= x^{4/6} + x^{7/6} + \dots, & y_4 &= x^{4/6} - x^{7/6} + \dots, \\ y_2 &= \omega x^{4/6} + x^{7/6} + \dots, & y_5 &= \omega x^{4/6} - x^{7/6} + \dots, \\ y_3 &= \omega^2 x^{4/6} + x^{7/6} + \dots, & y_6 &= \omega^2 x^{4/6} - x^{7/6} + \dots; \end{aligned}$$

hence calculating (what is most convenient) twice the number of intersections of the branch with itself, the partial branch y_1 intersects the other partial branches in $\frac{4}{6}, \frac{4}{6}, \frac{7}{6}, \frac{4}{6}, \frac{4}{6}$ points respectively, giving the sum $\frac{20}{6} + \frac{7}{6} = 4\frac{1}{2}$; each other partial branch intersects the remaining five branches in the same number of points; and therefore twice the number of intersections is $= 47$.

For the singularity $y = x^{4/6} + x^{7/6} + \dots$ in question, I say that if this be equivalent as above to δ' double points, κ' cusps, τ' double tangents and ι' inflexions, then that the number 47 just obtained is the value of $2\delta' + 3\kappa'$, and moreover, that the value of κ' is $\kappa' = \alpha - 1 = 5$; that is, we have $2\delta' + 3\kappa' = 47$ and $\kappa' = 5$; or, what is the same thing, $\delta' = 16$; $\kappa' = 5$. For the determination of the numbers τ', ι' , it is to be observed that the foregoing theory of branches is a theory of the points of a branch, by means of point-coordinates: there is a precisely similar theory of the tangents of a branch by means of line-coordinates, and we may inquire as to the number of the common tangents of two partial branches; and thence as to the number of common tangents of two branches, or of a branch with itself—it will appear that the line-equation of the branch is $Z = X^{p/(p-1)} \dots + X^{q/(q-1)} \dots$, so that the branch (which is as to its points sextic, $\alpha = 6$) is as

to its tangents quadric, $\beta = 2$, the two partial branches have with each other the number $= \frac{15}{2}$ of common tangents, or twice this number is $= 15$; that is, we have $2\tau' + 3\iota' = 15$, and moreover $\iota' = \beta - 1 = 1$, that is $\tau' = 6$, $\iota' = 1$; or finally for the singularity in question, the numbers $\delta', \kappa', \tau', \iota'$ are $= 16, 5, 6, 1$ respectively.

And so generally in the case of a branch which is α -ic, having with itself a number $= \frac{1}{2}M$ of common points; and as to its tangents β -ic, having with itself a number $= \frac{1}{2}N$ of common tangents, we have $2\delta' + 3\kappa' = M$, $\kappa' = \alpha - 1$, $2\tau' + 3\iota' = N$, $\iota' = \beta - 1$, or, what is the same thing, the values of $\delta', \kappa', \tau', \iota'$ are

$$\begin{aligned}\delta' &= \frac{1}{2}[M - 3(\alpha - 1)], \\ \kappa' &= \alpha - 1, \\ \tau' &= \frac{1}{2}[N - 3(\beta - 1)], \\ \iota' &= \beta - 1.\end{aligned}$$

I say that a singularity is simple when we have one branch, compound when we have more than one branch; the case above considered is that of a simple singularity, viz. we have on the curve one point, one tangent, one branch.

We may have a compound singularity where the branches all touch, that is we may have one point, one tangent, several branches. It may be seen that if $\frac{1}{2}M$ denote the number of common points of all the branches (that is of each branch with itself, and of every two branches with each other), and in like manner if $\frac{1}{2}N$ denote the number of common tangents of all the branches (that is of each branch with itself, and of every two branches with each other), then the formulae are

$$\begin{aligned}\delta' &= \frac{1}{2}[M - 3\Sigma(\alpha - 1)], \\ \kappa' &= \Sigma(\alpha - 1), \\ \tau' &= \frac{1}{2}[N - 3\Sigma(\beta - 1)], \\ \iota' &= \Sigma(\beta - 1),\end{aligned}$$

where the signs Σ refer to the different branches.

Again, we may have a compound singularity, one point, several tangents with to each of them a branch or branches; here if M denote the number of the common points of all the branches, and N the number of the common tangents of all the branches belonging to any one tangent, then the formulae are

$$\begin{aligned}\delta' &= \frac{1}{2}[M - 3\Sigma'\Sigma(\alpha - 1)], \\ \kappa' &= \Sigma'\Sigma(\alpha - 1), \\ \tau' &= \frac{1}{2}\Sigma'[N - 3\Sigma(\beta - 1)], \\ \iota' &= \Sigma'\Sigma(\beta - 1),\end{aligned}$$

where the signs Σ refer to all the branches belonging to the same tangent, and the signs Σ' to the different tangents. It is to be remarked, that the point on the curve is equivalent to the δ' double points and κ' cusps; each tangent is equivalent to $\frac{1}{2}[N - 3\Sigma(\beta - 1)]$ double tangents, and $\Sigma(\beta - 1)$ inflexions, the numbers N, β referring of course to the tangent in question.

Lastly, we may have a compound singularity, one tangent, several points (of contact), with to each of them a branch or branches; here if N denote the number of the common tangents of all the branches, M the number of the common points of all the branches belonging to any one point of contact, the formulae are

$$\begin{aligned}\delta' &= \frac{1}{2}\Sigma'[M - 3\Sigma(\alpha - 1)], \\ \kappa' &= \Sigma'\Sigma(\alpha - 1), \\ \tau' &= \frac{1}{2}[N - 3\Sigma'\Sigma(\beta - 1)], \\ \iota' &= \Sigma'\Sigma(\beta - 1),\end{aligned}$$

where the signs Σ refer to all the branches belonging to the same point of contact, and the signs Σ' to the different points of contact; it is to be remarked that the tangent of the curve is equivalent to the τ' double tangents and ι' inflexions; each point of contact is equivalent to $\frac{1}{2}[M - 3\Sigma(\alpha - 1)]$ double points and $\Sigma(\alpha - 1)$ cusps, the numbers M, α referring of course to the point of contact in question.

There is no difficulty in passing to the case of the compound singularity when the formulae for the simple singularity, one point, one tangent, one branch, are once obtained, and I now go back to the consideration of this case.

The class of a curve is equal to the number of tangents which can be drawn through an arbitrary point: the points of contact of these tangents are given as the intersections of the curve with a certain curve, the polar of the arbitrary point in regard to the curve; this polar passes through each double point and cusp, the double point counting as two points of intersection, and the cusp as three points of intersection (this is in fact the theory by which is found the reduction $= 2\delta + 3\kappa$ in the class of the curve). Hence, if the curve has a singularity $(\delta', \kappa', \tau', \iota')$, which to fix the ideas may be assumed to be a simple singularity, ‘one point, one tangent, one branch’; then the polar passes through the singular point, the number of intersections being $2\delta' + 3\kappa'$, or if the actual number of intersections be M , then we have $M = 2\delta' + 3\kappa'$. It is to be shown that the number M is equal to twice the number of common points which the curve has with itself at the singular point, so that the last-mentioned number is $= \frac{1}{2}M$. Suppose in the first instance that there is only a single branch, and let the branch be given by the equation

$$P = y + Ax^p + Bx^q + \dots = 0,$$

or introducing for homogeneity the third coordinate z , let this equation be

$$P = yz^{-1} + Ax^p z^{-p} + Bx^q z^{-q} + \dots = 0,$$

and let $P_1 = 0, P_2 = 0, \dots, P_\alpha = 0$, be the corresponding equations for the component partial branches; it is allowable to write $P_1 P_2 \dots P_\alpha = 0$ for the equation of the curve². Hence if (α, β, γ) be the coordinates of the arbitrary point, or putting in the first instance $\gamma = 1$, if $(\alpha, \beta, 1)$ be the coordinates, then writing $\Delta = \alpha\delta_x + \beta\delta_y + \delta_z$, the equation of the polar is $\Delta P_1 P_2 \dots P_\alpha = 0$, or, what is the same thing,

$$P_2 P_3 \dots P_\alpha \Delta P_1 + P_1 P_3 \dots P_\alpha \Delta P_2 + \text{etc.} = 0,$$

and we have

$$\Delta P = \alpha(pAx^{p-1}z^{-p} + qBx^{q-1}z^{-q} + \dots) + \beta z^{-1} - (pAx^p z^{-p-1} + qBx^q z^{-q-1} + \dots),$$

or putting $z = 1$, this is

$$\Delta P = \alpha(pAx^{p-1} + qBx^{q-1} + \dots) + \beta - (pAx^p + qBx^q + \dots),$$

and we have thence the values of $\Delta P_1, \Delta P_2 \dots \Delta P_\alpha$; the thing to be observed is, that the equation $\Delta P = 0$ is not satisfied (and therefore also each of the equations $\Delta P_1 = 0, \dots, \Delta P_\alpha = 0$ is not satisfied) by the coordinates $x = 0, y = 0$ of the singular point. We have now with the equation $\Delta P_1 P_2 \dots P_\alpha = 0$ of the polar to combine the equation $P_1 P_2 \dots P_\alpha = 0$: the last-mentioned equation breaks up into the equations $P_1 = 0, P_2 = 0, \dots, P_\alpha = 0$; and selecting for example the equation $P_1 = 0$, this gives the system $P_1 = 0, P_2 P_3 \dots P_\alpha \Delta P_1 = 0$, or since we require only the intersections at the singular point, and $\Delta P_1 = 0$ does not pass through this point, this may be replaced by $P_1 = 0, P_2 P_3 \dots P_\alpha = 0$. The complete system is thus $(P_1 = 0, P_2 P_3 \dots P_\alpha = 0), (P_2 = 0, P_1 P_3 \dots P_\alpha = 0), \dots, (P_\alpha = 0, P_1 P_2 \dots P_{\alpha-1} = 0)$; or, what is the same thing, we have each pair $(P_r = 0, P_s = 0)$ taken twice. To eliminate y from these equations, we have merely to write $P_r - P_s = 0$, or, what is the same thing, we have $\psi(P_1, P_2 \dots P_\alpha) = 0$, ψ denoting the product of the squares of the differences of the functions $(P_1, P_2 \dots P_\alpha)$.

Suppose that any two partial branches $P_r = 0, P_s = 0$ intersect (according to the above-mentioned definition) in p points; then $P_r - P_s$ contains the factor x^p , and hence the product $\psi(P_1, P_2 \dots P_\alpha)$ contains as a factor x to the power $2\Sigma p$, that is, the equation in x has $2\Sigma p$ roots each $= 0$. Whence if $2\Sigma p = \frac{1}{2}M$, then the equation in x has M roots each $= 0$, or the curve and polar have at the singular point M intersections, that is $M = 2\delta' + 3\kappa'$.

I have no complete proof to offer of the remaining equation $\kappa' = \alpha - 1$, it was obtained from the consideration of a particular case as follows. Consider the linear branch $y = Ax^p + \dots$, where the exponents are all positive integers, and taking the axis of x to be the tangent, the least exponent p is greater than

²Of course this is not the equation in its rational and integral form, and on this account the reasoning of the text is not free from difficulty; the same remark applies to a subsequent equation.

unity; if $p = 2$ there is at the origin no inflexion, if $p = 3$ there is a single inflexion, and generally the number of inflexions is $= p - 2$. Now it will presently appear that in line-coordinates the equation of the branch is $Z = A'X^{p/(p-1)}$, or replacing Z, X by the original point-coordinates y, x the branch $y = A'x^{p/(p-1)} + \dots$ has at the origin $p - 2$ cusps; but in the branch in question we have $\alpha = p - 1$, and the number of cusps is thus $= \alpha - 1$; this result is confirmed by other particular instances, and I assume in general that we have $\kappa' = \alpha - 1$; whence in the case of a simple singularity, or where there is only one branch we have $M = 2\delta' + 3\kappa', \kappa' = \alpha - 1$, or, what is the same thing, $\delta' = \frac{1}{2}[M - 3(\alpha - 1)], \kappa' = \alpha - 1$. The reasoning is easily adapted to the case of a compound singularity.

I consider the branch

$$y + Ax^p + Bx^q + \dots = 0,$$

(where it is assumed that the axis of x is a tangent to the branch, and therefore that the lowest exponent p is greater than unity), introducing the coordinate z for homogeneity, this becomes

$$yz^{-1} + Ax^pz^{-p} + Bx^qz^{-q} + \dots = 0,$$

and I proceed to find the corresponding equation in line coordinates, taking these to be X, Y, Z , we have

$$\begin{aligned}\lambda X &= pAx^{p-1}z^{-p} + qBx^{q-1}z^{-q} + \dots, \\ \lambda Y &= z^{-1}, \\ \lambda Z &= -yz^{-2} - pAx^pz^{-p-1} - qBx^qz^{-q-1} + \dots,\end{aligned}$$

or writing $z = 1, Y = 1$, we find $\lambda = 1$, and therefore

$$\begin{aligned}X &= pAx^{p-1} + qBx^{q-1} + \dots, \\ Z &= -y - pAx^p - qBx^q + \dots,\end{aligned}$$

here substituting for $-y$ its value $= Ax^p + Bx^q + \dots$, we have

$$\begin{aligned}X &= pAx^{p-1} + qBx^{q-1} + \dots, \\ Z &= (1 - p)Ax^p + (1 - q)Bx^q + \dots\end{aligned}$$

Hence writing $pAx^{p-1} = \theta$, the equations are

$$\begin{aligned}X &= \theta - B'\theta^{q-1} - \dots, \\ Z &= -A'\theta^{p/(p-1)} - B'\theta^{q/(p-1)} - \dots,\end{aligned}$$

so that eliminating θ , we have

$$Z = A'X^{p/(p-1)} + B'X^{q/(p-1)} + \dots,$$

and it is easy to see by Lagrange's theorem, that the general form of the exponents in the series on the right-hand side is $\frac{p+f(q-p)+g(r-p)+\dots}{p-1}$, where $f, g, \dots$ are positive integers, zero included. The equation in line-coordinates being known, the subsequent investigation is precisely the same as that for the point-coordinates, and hence in the case of one branch, if this be in regard to its tangents β -ic, and have $\frac{1}{2}N$ common tangents with itself, then $2\tau' + 3\iota' = N, \iota' = \beta - 1$, or, what is the same thing, $\tau' = \frac{1}{2}[N - 3(\beta - 1)], \iota' = \beta - 1$. The investigation in the case of a simple singularity of the values of $\delta', \kappa', \tau', \iota'$ is thus completed.

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NOTES ON POLYHEDRA.

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Axial Properties. Article 1 to 18.

1. A POLYHEDRON may have a q -axis, viz. a line about which if it is made to rotate through an angle $= \frac{2\pi}{q}$ (but not through any sub-multiple of this angle), it will occupy the same portion of space. It is then clear that when the rotation is repeated any number of times the body will still occupy the same portion of

space; or if Θ denote the rotation through the angle $\frac{2\pi}{q}$, then we have the rotations $1, \Theta, \Theta^2, \dots, \Theta^{q-1}$, and finally $\Theta^q = 1$, that is, when the rotation is q -times repeated, the body will resume its original position. Similarly for any number of axes ($\Theta^q = 1, \Theta'^{q'} = 1, \dots$, where the indices $q, q', \dots$ may be the same or different) we have the rotations $1, \Theta, \Theta^2, \dots, \Theta^{q-1}, \Theta', \Theta'^2, \dots, \Theta'^{q'-1}, \dots$; and if $\Theta, \Theta', \dots$ be the entire system of the axes of the body, these rotations will form a group. The rotations in question are in fact the entire series of those which leave unaltered the portion of space occupied by the body, and since any two rotations combine together into a single rotation, any two of the rotations in question must combine together into some one of these rotations, that is, the rotations in question form a group. Some analytical consequences of this theorem will be obtained in the sequel.

2. The number of axes may be denoted by $\Sigma 1$ and the number of rotations by $1 + \Sigma(q - 1)$; we may say that $\Sigma 1$ is the number, and $1 + \Sigma(q - 1)$ the efficiency or weight, of the axes.

3. For any one of the regular polyhedra, E being the number of edges, then the number of axes or $\Sigma 1$ is $= \frac{1}{2}E + 1$, and their weight or $1 + \Sigma(q - 1)$ is $= 2E$. In fact, if as usual S denote the number of summits, F the number of faces, and if there be m edges to a face, and n edges to a summit, then $S + F = E + 2$, $mF = nS = 2E$. Now in all the polyhedra except the tetrahedron, we have a number $\frac{1}{2}F$ of m -axes passing through the centres of opposite faces (amphihedral axes as Mr Kirkman has termed them) and a number $\frac{1}{2}S$ of n -axes passing through opposite summits (amphigonal axes); and we have besides a number $\frac{1}{2}E$ of 2-axes passing through the mid-points of opposite edges (amphigrammic axes): the entire number of axes is thus $\frac{1}{2}(S + F + E)$, which is $= E + 1$; and the weight is $1 + \frac{1}{2}F(m - 1) + \frac{1}{2}S(n - 1) + \frac{1}{2}E$, which is $= 1 + \frac{1}{2}mF + \frac{1}{2}nS - \frac{1}{2}(F + S - E) = 1 + E + E - 1 = 2E$. In the case of the tetrahedron $S = F = 4, m = n = 3$, and the only difference is that instead of the $\frac{1}{2}F$ amphihedral m -axes and the $\frac{1}{2}S$ amphigonal n -axes, we have a number $(F = S) = \frac{1}{2}(F + S)$ of $(n = m)$ -gonal axes each through a summit and the centre of an opposite face (gonohedral axes).

4. The theorem that the weight $1 + \Sigma(q - 1) = 2E$, or say $1 + \Sigma(q - 1) = mF$, may be extended so as to apply to any polyhedron whatever. In fact considering any face A of the polyhedron, let F be the number of faces homologous to (and inclusive of) A ; and, taking a any edge of the face A , let m be the number of edges of A homologous to (and inclusive of) a : then we have $1 + \Sigma(q - 1) = mF$. This is almost a truism when the signification of the term “homologous” is explained. Imagine the polyhedron placed on a plane, say the table, and draw on the table a polygon equal to the polygonal face A , and in this polygon select some one edge corresponding to the edge a . The polyhedron may be placed on the table with the face A coinciding with the polygon, or say the face A may be superimposed on the polygon, and that in m different ways, viz. any one of the edges homologous to a may be made to coincide with the assumed edge: and in like manner there are F different faces (viz. the faces homologous to A) which may be superimposed on the polygon, each of them in m different ways; that is there are in all mF different positions of the polyhedron for each of which it occupies the same portion of space. And we have thus the required theorem $1 + \Sigma(q - 1) = mF$.

5. As an example, take the regular pyramid on a square base; there is here a single axis, viz. a 4-axis, and we have $1 + \Sigma(q - 1) = 1 + 3 = 4$. If for the face A we take the square base, then there is no other face homologous thereto and therefore $F = 1$; but the four sides are homologous to each other or $m = 4$, and we have $mF = 4$. Similarly taking for A one of the triangular faces, since these are homologous to each other, then $F = 4$; and if we take for the side a the base of the triangle, then there is no other side homologous to this, or $m = 1$; and therefore $mF = 4$. It might at first sight appear that the two equal sides of the triangle were homologous to each other, and therefore that taking for the edge a one of these sides we should have $m = 2$; but in fact although the two sides in question are homologously related to the pyramid, yet according to the definition they are not homologous sides of the triangular face, and we still have $m = 1$, and therefore $mF = 4$.

6. Of course in the case of a regular polyhedron the faces are all homologous, and the edges of a face are all homologous, that is F will denote the entire number of faces, and m the number of edges to a face: so that for this case the theorem gives $1 + \Sigma(q - 1) = 2E$ as above.

7. Returning to the regular polyhedra the axial systems are

Tetrahedron	$4L^3, 3L^2$
Cube and Octahedron	$3L^4, 4L^3, 6L^2$
Dodecahedron and Icosahedron	$6L^5, 10L^3, 15L^2$

where L^3 denotes a 3-axis, etc.; this is in accordance with the notation of M. Bravais in the memoir

subsequently referred to.

8. The regular polyhedra may be exhibited in connexion with each other as follows: Imagine the polyhedron projected on a concentric sphere by lines through the centre; so that the summits become points on the sphere, the edges arcs of great circles, and the faces spherical polygons. Starting from the dodecahedron, the centres of the pentagonal faces are the summits of the icosahedron, and conversely for the icosahedron the centres of the triangular faces are the summits of the dodecahedron: moreover each edge of the dodecahedron cuts at right angles an edge of the icosahedron and the two edges have the same mid-point. Again if in any face of the dodecahedron we draw one of the five diagonals (arcs through two non-adjacent summits) there is in the face a single edge not met by this diagonal; and in the other face through this edge a single diagonal not met by the edge; joining the extremities of the two diagonals we have a spherical square, the face of the cube; it is to be observed that the summits of the cube are eight out of the twenty summits of the dodecahedron, and that the centres of the faces of the cube are the mid-points of six out of the thirty edges of the dodecahedron or the icosahedron. The cube given by the foregoing construction is of course one out of five different cubes. The centres of the faces of the cube are the summits of the octahedron; and conversely the centres of the faces of the octahedron are the summits of the cube; moreover each edge of the cube cuts at right angles an edge of the octahedron; and the two edges have the same mid-point. Finally, taking four non-adjacent summits of the cube (which can be done in two different ways), these are the summits of the tetrahedron, and the mid-points of the edges of the tetrahedron are the summits of the octahedron.

9. Considering the polyhedra in the foregoing mutual connexion, all the axes of the tetrahedron are axes of the cube and octahedron, viz. the 2-axes of the tetrahedron are the 4-axes of the cube and octahedron; and the 3-axes of the tetrahedron are the 3-axes of the cube and octahedron; moreover the 3-axes of the cube and octahedron are included among the 3-axes of the dodecahedron and icosahedron and the 4-axes of the cube and octahedron are included among the 2-axes of the dodecahedron and icosahedron; but the 2-axes of the cube and octahedron are not included among the axes of the dodecahedron and icosahedron. The 4-axes of the cube and octahedron form thus a system of rectangular axes common to all the polyhedra, and representing these axes (or say the summits of the corresponding rectangular spherical triangle) by X, Y, Z , we have a convenient system of coordinate axes to which to refer all the other axes of the polyhedron, viz. if P be the extremity (chosen at pleasure) of the axis in question, then the position of the axis may be determined by its distance PZ and azimuth XPZ (measured in the direction from X to Y), or by its distances PX, PY, PZ , or say X, Y, Z from the three rectangular axes (we have, it is clear, $\cos X = \sin \text{dist.} \cos \text{azim.}$, $\cos Y = \sin \text{dist.} \sin \text{azim.}$, $\cos Z = \cos \text{dist.}$). The rotation angle of a q -axis is $= \frac{2\pi}{q}$ (i.e. this is the angle through which if the body be turned about the axis, it still occupies the same portion of space) and the half-rotation angle is therefore $= \frac{\pi}{q}$. Moreover if i, j, k are Sir W. R. Hamilton's quaternion symbols, then the "rotation symbol" of the axis is

$$\cos \frac{\pi}{q} + \sin \frac{\pi}{q}(i \cos X + j \cos Y + k \cos Z),$$

the application of which will be presently explained.

10. The angular coordinates of the different axes may be found by spherical trigonometry without much difficulty; and we are then able to form the following axial tables of the several polyhedra: the extremity of each axis is chosen in such manner that the distance PZ is not $> 90^\circ$.

Axial System of the Tetrahedron.

4 3-axes, $\frac{1}{2}$ **Rot. angle** $= 60^\circ$, $\cos = \frac{1}{2}$, $\sin = \frac{1}{2}\sqrt{3}$.

Distance		Azimuth		$\cos X$	$\cos Y$	$\cos Z$
angle	cos	angle	$\cos / \sin$			
$54^\circ 44'$	$\frac{1}{\sqrt{3}}$	45°	$\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$
"	"	135°	$-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$
"	"	225°	$-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$
"	"	315°	$\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$

Rot. Symbols: $\frac{1}{2}(1 + i + j + k)$, $\frac{1}{2}(1 - i + j + k)$, $\frac{1}{2}(1 - i - j + k)$, $\frac{1}{2}(1 + i - j + k)$.

3 2-axes, $\frac{1}{2}$ **Rot. angle** $= 90^\circ$, $\cos = 0$, $\sin = 1$.

angle	cos	angle	cos / sin	cos X	cos Y	cos Z	Rot. Symbol
90°	0	0°	1, 0	1	0	0	i
90°	0	90°	0, 1	0	1	0	j
0°	1	—	—	0	0	1	k

Axial System of the Cube and Octahedron.

4 3-axes, $\frac{1}{2}$ **Rot. angle** = 60°, cos = $\frac{1}{2}$, sin = $\frac{1}{2}\sqrt{3}$.

Distance		Azimuth		cos X	cos Y	cos Z	Rot. Symbol
angle	cos	angle	cos / sin				
54°44'	$\frac{1}{\sqrt{3}}$	45°	$\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{2}(1 + i + j + k)$
”	”	135°	—	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{2}(1 - i + j + k)$
”	”	225°	—	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{2}(1 - i - j + k)$
”	”	315°	—	$\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{2}(1 + i - j + k)$

3 4-axes, $\frac{1}{2}$ **Rot. angle** = 45°, cos = $\frac{1}{\sqrt{2}}$, sin = $\frac{1}{\sqrt{2}}$.

angle	cos	angle	cos / sin	cos X	cos Y	cos Z	Rot. Symbol
0°	1	—	—	0	0	1	$\frac{1}{\sqrt{2}}(1 + k)$
90°	0	0°	1, 0	1	0	0	$\frac{1}{\sqrt{2}}(1 + i)$
90°	0	90°	0, 1	0	1	0	$\frac{1}{\sqrt{2}}(1 + j)$

6 2-axes, $\frac{1}{2}$ **Rot. angle** = 90°, cos = 0, sin = 1.

angle	cos	angle	cos / sin	cos X	cos Y	cos Z	Rot. Symbol
45°	$\frac{1}{\sqrt{2}}$	0°	1, 0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}(i + k)$
”	”	90°	0, 1	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}(j + k)$
”	”	180°	-1, 0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}(-i + k)$
”	”	270°	0, -1	0	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}(-j + k)$
90°	0	45°	$\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}(i + j)$
”	”	135°	$-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}(-i + j)$

Axial System of the Dodecahedron and Icosahedron.

[Figure: extensive tabular listing across two pages (pp. 534–535 in original) of the 6 5-axes (half-rotation angle 36°, cos = $\frac{\sqrt{5}+1}{4}$, sin = $\frac{1}{4}\sqrt{10-2\sqrt{5}}$), 10 3-axes (half-rotation 60°), and 15 2-axes (half-rotation 90°) with their distance, azimuth, direction cosines cos X , cos Y , cos Z , and quaternion rotation symbols built from i, j, k and the golden-ratio expressions $\frac{\sqrt{5}\pm 1}{4}$. The data tabulates, for each axis, the chosen extremity together with its three direction cosines and the corresponding rotation symbol $\cos \frac{\pi}{q} + \sin \frac{\pi}{q}(i \cos X + j \cos Y + k \cos Z)$.]

11. Before proceeding further I remark that exclusively of the foregoing axial systems of the regular polyhedra the only cases are as follows:

A. A polyhedron may have a single q -axis, say A^q : taking this as the axis of Z the table is

X	Y	Z	Rot. Symbol
90°	90°	0°	$\cos \frac{\pi}{q} + \sin \frac{\pi}{q} \cdot k$

B. It may have a single q -axis, and (symmetrically arranged in a plane at right angles thereto) q 2-axes, say A^q, qD . Taking the q -axis as the axis of Z and some one of the 2-axes as the axis of X , the table is

X	Y	Z	Rot. Symbols
One q -axis, $\frac{1}{2}$ Rot. angle $= \frac{\pi}{q}$.			
90°	90°	0°	$\cos \frac{\pi}{q} + \sin \frac{\pi}{q} \cdot k$
q 2-axes, $\frac{1}{2}$ Rot. angle $= 90^\circ$.			
0°	90°	90°	i
$\frac{\pi}{q}$	$90^\circ - \frac{\pi}{q}$	90°	$i \cos \frac{\pi}{q} + j \sin \frac{\pi}{q}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(q-1)\frac{\pi}{q}$	$90^\circ - \frac{(q-1)\pi}{q}$	90°	$i \cos \frac{(q-1)\pi}{q} + j \sin \frac{(q-1)\pi}{q}$

and in particular if $q = 2$, the axes are $3L^2$ and the table is

X	Y	Z	Rot. Symbols
3 2-axes, $\frac{1}{2}$ Rot. angle $= 90^\circ$.			
90°	90°	0°	k
0°	90°	90°	i
90°	0°	90°	j

This in fact appears, Bravais, “Mémoire sur les polyèdres de forme symétrique,” *Liouville*, t. xiv., pp. 141–180 (1849), observing that for the present purpose there is no distinction between his three cases

$$A^{2q+1}, (2q+1)L^2; \quad A^{2q}, qL^2, qL'^2; \quad A^{2q}, 2qD.$$

12. The meaning of the rotation symbol is as follows: viz. if in general we have a rotation θ about an axis inclined at the angles X, Y, Z to any three rectangular axes, and if Π be the rotation symbol,

$$\Pi = \cos \frac{1}{2}\theta + \sin \frac{1}{2}\theta(i \cos X + j \cos Y + k \cos Z),$$

then if x, y, z are the original coordinates of any point of the body, and x', y', z' the coordinates of the same point after the rotation; the values of x', y', z' are given in terms of x, y, z by the formula

$$ix' + jy' + kz' = \Pi(ix + jy + kz)\Pi^{-1}.$$

This is in fact the form under which, in the paper “On certain results relating to Quaternions,” *Phil. Mag.*, vol. xxvi. (1845), p. 141, [20], I exhibited the rotation formulae of Euler and Rodrigues. See also my paper “On the application of Quaternions to the Theory of Rotation,” *Phil. Mag.*, vol. xxxiii. (1848), p. 196, [68].

13. We have, it is clear,

$$\Pi^s = \cos \frac{s\theta}{2} + \sin \frac{s\theta}{2}(i \cos X + j \cos Y + k \cos Z)$$

which shows that Π^s is the symbol for the rotation Π repeated s times: (more generally performing on the body, first the rotation Π and then the rotation Φ about any axis, the same or different, the symbol of the resultant rotation is $= \Phi\Pi$). If Π be the symbol for a rotation through the angle $\frac{2\pi}{q}$, then the rotation which corresponds to the symbol Π^q is a rotation through 360° , that is the body returns to its original position; it might at first sight appear that we ought to have $\Pi^q = 1$, and that the symbols $1, \Pi, \Pi^2, \dots, \Pi^{q-1}$ would form a group; this however is not so, for we have not $\Pi^q = 1$, but $\Pi^q = -1$; in fact, it is to be observed that to pass from $ix + jy + kz$ to $ix' + jy' + kz'$, we have to multiply by $\Pi () \Pi^{-1}$, so that the symbol of the rotation is indifferently $\pm\Pi$, and that the rotation symbol -1 is thus equivalent to the rotation symbol $+1$. But as regards the formation of the group, the only difference is that it is not $1, \Pi, \Pi^2, \dots, \Pi^{q-1}$ which form a group of q symbols, but $+1, \pm\Pi, \pm\Pi^2, \dots, \pm\Pi^{q-1}$ which form a group of $2q$ symbols. And so in the axial system of any polyhedron, if Π be the rotation symbol of any q -axis, then taking for each axis of the polyhedron the set of symbols $\pm\Pi, \pm\Pi^2, \dots, \pm\Pi^{q-1}$, and besides the two symbols ± 1 , the whole series of symbols form together a group.

14. Thus in the before-mentioned case $B(q = 2)$ we have the eight symbols

$$\pm 1, \pm i, \pm j, \pm k$$

forming (as they obviously do) a group. In the general case B , putting for shortness

$$\Theta_s = \cos \frac{s\pi}{q} + \sin \frac{s\pi}{q} \cdot k \quad \text{and} \quad \Phi_s = i \cos \frac{s\pi}{q} + j \sin \frac{s\pi}{q},$$

the group consists of the $4q$ symbols

$$\pm 1, \pm \Theta_1, \dots \pm \Theta_{q-1}; \pm \Phi_0, \pm \Phi_1, \dots \pm \Phi_{q-1},$$

(to verify that this is so, it is only necessary to form the equations

$$\Theta_s \Theta_t = \Theta_{s+t}, \quad \Phi_s \Phi_t = -\Theta_{s-t}, \quad \Theta_s \Phi_t = \Phi_{s+t}, \quad \Phi_s \Theta_t = \Phi_{s-t},$$

which are at once seen to be true).

15. The general case A gives merely the group of the $2q$ symbols

$$\pm 1, \pm \Theta_1, \dots \pm \Theta_{q-1},$$

which has been already mentioned.

16. The tetrahedron gives the group of 24 symbols,

$\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$	16 cube roots of +1
$\pm i, \pm j, \pm k$	6 square roots „ „
± 1	2 terms
	24

(the signs $\pm$ being all independent).

17. The cube and octahedron give the group of 48 symbols

$\frac{1}{\sqrt{2}}(\pm 1 \pm i), \frac{1}{\sqrt{2}}(\pm 1 \pm j), \frac{1}{\sqrt{2}}(\pm 1 \pm k)$	12 fourth roots of ± 1
$\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$	16 cube „ „ „
$\pm i, \pm j, \pm k, \frac{1}{\sqrt{2}}(\pm j \pm k), \frac{1}{\sqrt{2}}(\pm k \pm i), \frac{1}{\sqrt{2}}(\pm i \pm j)$	18 square „ „ „
± 1	2 terms
	48

(the signs $\pm$ being all independent).

18. The dodecahedron and icosahedron give the group of 120 symbols

$\pm \frac{\sqrt{5}+1}{4} \pm \frac{1}{2}j \pm \frac{\sqrt{5}-1}{4}k,$	48 fifth roots of +1
$\pm \frac{\sqrt{5}+1}{4} \pm \frac{1}{2}k \pm \frac{\sqrt{5}-1}{4}i,$	
$\pm \frac{\sqrt{5}+1}{4} \pm \frac{1}{2}i \pm \frac{\sqrt{5}-1}{4}j,$	
$\frac{1}{2}(\pm 1 \pm i \pm j \pm k),$	
$\pm \frac{1}{2} \pm \frac{\sqrt{5}-1}{4}j \pm \frac{\sqrt{5}+1}{4}k,$	40 cube roots of ± 1
$\pm \frac{1}{2} \pm \frac{\sqrt{5}-1}{4}k \pm \frac{\sqrt{5}+1}{4}i,$	
$\pm \frac{1}{2} \pm \frac{\sqrt{5}-1}{4}i \pm \frac{\sqrt{5}+1}{4}j,$	
$\pm i, \pm j, \pm k,$	30 square „ „
$\pm \frac{\sqrt{5}+1}{4}j \pm \frac{\sqrt{5}-1}{4}k,$	
$\pm \frac{\sqrt{5}+1}{4}k \pm \frac{\sqrt{5}-1}{4}i,$	
$\pm \frac{\sqrt{5}+1}{4}i \pm \frac{\sqrt{5}-1}{4}j,$	
± 1	2 terms

giving in all 120 symbols.

(The signs $\pm$ are all independent, except that in each of the three expressions in the top line the signs in $\sqrt{5}+1, \sqrt{5}-1$, are opposite to each other, so that each of the three expressions has 16 values.)

It is to be remarked that in the groups of 24 and 48, the group is not altered by any permutation whatever of the symbols i, j, k ; whereas the group of 120 is not altered by the cyclical permutation of these symbols, but it is altered by the interchange of any two of them; the geometrical reason of this difference may be perceived without difficulty.

P.S. I found accidentally, *Gergonne*, t. xv., p. 40, (1824–25), the following problem: “De combien de manières m couleurs différentes les unes des autres peuvent-elles être appliquées sur les faces d’un polyèdre régulier; m représentant tour à tour les nombres 4, 6, 8, 12, 20?”

Instead of the m of the problem, writing as before F for the number of faces, and writing also E for the number of edges; then if different positions of the same polyhedron were reckoned as different polyhedra, the number of ways would of course be $\Pi(F)$ ($= 1 \cdot 2 \cdot 3 \dots F$); and since by what precedes the same polyhedron can be placed in $2E$ positions, the required number of ways is $\frac{1}{2E}\Pi(F)$.

Thus for the tetrahedron, if the colours are black, white, red, green, we may place it with the black face on the table and the white face in front; the only variation in the disposition of the colours, is according as the right hand and the left hand faces are coloured red and green or else green and red respectively; and the number of ways therefore is $= 2$, which agrees with the formula.

2, Stone Buildings, W.C., 30 January, 1863.

376.

THÉORÈME RELATIF À L'ÉQUILIBRE DE QUATRE FORCES.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. lxi. (*Juillet–Décembre* 1865), pp. 829–830.]

On sait que si quatre forces qui agissent sur un corps solide se tiennent en équilibre, alors (théorème de M. Möbius) les droites suivant lesquelles ces forces agissent sont quatre génératrices d’un même hyperboloïde: et de plus en représentant chaque force par une longueur proportionnelle sur la direction de cette force, alors (théorème de M. Chasles) le tétraèdre formé par deux quelconques des forces est égal au tétraèdre formé par les deux autres forces.

En cherchant les valeurs des quatre forces lesquelles en agissant selon quatre génératrices données d’un même hyperboloïde se tiennent en équilibre, j’ai réussi à trouver pour ces valeurs une expression assez remarquable qui comprend comme corollaire le théorème de M. Chasles.

Je nomme moment de deux droites la distance perpendiculaire de ces droites multipliée par le sinus de leur inclinaison mutuelle. Cela étant, en considérant quatre droites 1, 2, 3, 4 génératrices d’un même hyperboloïde, je dénote par ces mêmes symboles 1, 2, 3, 4 les forces qui agissent selon ces quatre droites respectivement, et par 12 le moment des droites 1 et 2, et de même pour les autres combinaisons de deux droites.

Or je dis que les forces 1, 2, 3, 4 qui se tiennent en équilibre ont les valeurs proportionnelles que voici, à savoir en prenant les radicaux avec des signes convenables:

$$\begin{aligned} 1 &= \sqrt{23 \cdot 34 \cdot 42}, \\ 2 &= \sqrt{34 \cdot 41 \cdot 13}, \\ 3 &= \sqrt{41 \cdot 12 \cdot 24}, \\ 4 &= \sqrt{12 \cdot 23 \cdot 31}. \end{aligned}$$

On déduit de là, en écrivant pour abrégé

$$\nabla = \sqrt{23 \cdot 31 \cdot 12 \cdot 14 \cdot 24 \cdot 34},$$

les équations

$$\begin{aligned} \nabla \sqrt{23 \cdot 14} &= 2 \cdot 3 \cdot 23 = 1 \cdot 4 \cdot 14, \\ \nabla \sqrt{31 \cdot 24} &= 3 \cdot 1 \cdot 31 = 2 \cdot 4 \cdot 24, \\ \nabla \sqrt{12 \cdot 34} &= 1 \cdot 2 \cdot 12 = 3 \cdot 4 \cdot 34, \end{aligned}$$

où par exemple l’équation $1 \cdot 2 \cdot 12 = 3 \cdot 4 \cdot 34$ exprime que le produit des forces 1 et 2 par le moment 12 des droites suivant lesquelles ces forces agissent est égal au produit des forces 3 et 4 par le moment 34 des droites selon lesquelles ces forces agissent.

J'ajoute que l'on a, en prenant les radicaux avec les signes convenables,

$$\sqrt{23 \cdot 14} + \sqrt{31 \cdot 24} + \sqrt{12 \cdot 34} = 0,$$

équation qui subsiste non seulement pour quatre génératrices quelconques d'un même hyperboloïde, mais pour quatre droites liées par une relation géométrique plus générale, à savoir, pour quatre droites telles que les deux droites qui rencontrent ces quatre droites se réduisent à une seule droite: ou (ce qui est la même chose) telles que chacune des quatre droites touche l'hyperboloïde qui passe par les trois autres droites.

377.

NOTE SUR LA CORRESPONDANCE DE DEUX POINTS SUR UNE COURBE.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. lxii. (Janvier–Juin, 1866), pp. 586–590.]

Dans la théorie à laquelle se rapporte cette Note, un point de rebroussement, s'il était nécessaire d'en parler, serait censé un cas particulier du point double; mais, pour simplifier, je ne ferai attention qu'aux courbes sans point de rebroussement.

Une courbe de l'ordre m peut avoir au plus $\frac{1}{2}(m-1)(m-2)$ points doubles; la différence entre ce nombre et le nombre actuel δ des points doubles d'une courbe donnée, savoir le nombre

$$D = \frac{1}{2}(m-1)(m-2) - \delta,$$

que je nomme le défaut (en anglais, *deficiency*), joue, comme on sait, un rôle important dans la théorie de la courbe. En particulier, pour une courbe de l'ordre m avec le défaut $D = 0$, ou, comme je dis, pour une courbe unicursale de l'ordre m , les coordonnées (x, y, z) d'un point quelconque de la courbe (je me sers toujours des coordonnées homogènes) sont proportionnelles à des fonctions rationnelles et entières du degré m d'un paramètre variable θ .

Cela étant, le théorème de M. Chasles: "Lorsque sur une droite deux séries de points P, P' se correspondent de manière qu'à un point donné P correspondent α points P' , et qu'à un point donné P' correspondent α' points P , alors le nombre des points P qui coïncident avec les points correspondants P' est $\alpha + \alpha'$;" ce théorème, dis-je, s'étend sans changement à des points correspondants situés sur une courbe unicursale quelconque; et l'on peut énoncer le théorème comme il suit:

Lorsque, sur une courbe unicursale, il y a deux séries de points qui ont une correspondance (α, α') , le nombre des points unis est $\alpha + \alpha'$.

Cela donne lieu au théorème: "Lorsque, sur une courbe, avec le défaut D , il y a deux séries de points qui ont une correspondance (α, α') , le nombre des points unis est $\alpha + \alpha' + 2kD$," où $2k$ est un facteur qu'il s'agit de déterminer. Cela peut se faire, sinon toujours, au moins dans la plupart des cas, au moyen du théorème que voici, tiré d'une induction qui me paraît suffisante:

En considérant sur la courbe $U = 0$ un point donné P' , et puis les intersections de la courbe $U = 0$ par une courbe $\Theta = 0$ dont l'équation contient d'une manière quelconque les coordonnées (x', y', z') du point donné P' ; s'il y a k intersections qui coïncident avec le point P' , et que les autres intersections forment un système de points P qui correspondent au point donné P' , et si cette correspondance est une correspondance (α, α') , alors le nombre des points unis est $\alpha + \alpha' + 2kD$.

Je donne quatre exemples de ce théorème:

1°. *Recherche de la classe*.—Si les points correspondants P, P' sont situés en ligne droite avec un point donné O , alors les points unis sont les points de contact des tangentes menées par le point O ; donc le nombre des points unis est égal à la classe de la courbe. La courbe $\Theta = 0$ est ici la droite OP' , il y a donc une seule intersection P' ; donc $k = 1$, et nous avons entre les points P, P' une correspondance $(m-1, m-1)$. Donc nous avons pour la classe M de la courbe l'expression

$$M = 2(m-1) + 2D,$$

où, en substituant pour D la valeur

$$D = \frac{1}{2}(m-1)(m-2) - \delta,$$

nous trouvons

$$M = m^2 - m - 2\delta,$$

comme cela doit être.

2°. *Recherche du nombre des inflexions.*—Si les points P sont les points de rencontre avec la courbe de la tangente au point P' , alors les points unis seront les points d'inflexion. La courbe $\Theta = 0$ est ici la tangente au point P' ; il y a ainsi deux intersections au point P' ; donc $k = 2$; de plus, à chaque point P' correspondent $(m - 2)$ points P , et à chaque point P correspondent $M - 2$ points P' . On a donc pour le nombre des inflexions

$$\iota = (m + M - 4) + 4D,$$

ou, en substituant pour M, D , leurs valeurs,

$$\iota = 3m(m - 2) - 6\delta,$$

ce qui est juste.

Avant d'aller plus loin, il convient de généraliser le théorème, en remarquant que les intersections des courbes $U = 0, \Theta = 0$ peuvent former plusieurs systèmes simples ou multiples de points: les intersections peuvent être le point P' (k fois), un système de points P (p fois), un système de points Q (q fois), etc. Cela étant, s'il y a entre les points P' et P une correspondance (α, α') , et si le nombre des points unis de ce système est a ; s'il y a entre les points P' et Q une correspondance (β, β') , et si le nombre des points unis de ce système est b , et ainsi de suite; alors le théorème prend la forme

$$pa + qb + \dots = p(\alpha + \alpha') + q(\beta + \beta') + \dots + 2kD;$$

c'est la forme applicable à l'exemple qui suit.

3°. *Recherche du nombre des tangentes doubles.*—Prenons pour la courbe $\Theta = 0$ le système des $(M - 2)$ tangentes menées à la courbe par le point donné P' ; on a ici les points P qui sont les points de contact de ces tangentes, et les points Q qui sont les autres intersections de la courbe par ces tangentes; les intersections sont le point P' $(M - 2)$ fois (donc $k = M - 2$), le système des points P (2 fois) et le système des points Q (1 fois). Le système P, P' est précisément celui qui donne les points d'inflexion. On a donc

$$\alpha = \alpha' = m - 1;$$

a est égal au nombre de points d'inflexion (mais, pour plus de commodité, je retiens le symbole a); $p = 2$. Le système P, Q est un système qui a pour points unis les points de contact des tangentes doubles, le nombre b des points unis sera donc 2τ , en dénotant par τ le nombre des tangentes doubles. On a pour la correspondance (β, β') entre les points P' et Q

$$\beta = \beta' = (m - 3)(M - 2);$$

enfin

$$q = 1.$$

Le théorème donne ainsi

$$2a + b = 2(m + M - 4) + 2(2m - 3)(M - 2) + 2(M - 2)D;$$

mais nous avons ci-dessus trouvé

$$a = (m + M - 4) + 4D;$$

donc enfin

$$b = 2\tau = 2(m - 3)(M - 2) + 2(M - 6)D,$$

où, en substituant pour M et D leurs valeurs, on retrouve la formule ordinaire

$$2\tau = m(m - 2)(m^2 - 9) - (m^2 - m - 6) \cdot 4\delta + 4\delta(\delta - 1).$$

Parmi les intersections des courbes $U = 0, \Theta = 0$, il peut y avoir un système simple ou multiple de points fixes, c'est-à-dire indépendants de la position du point P' ; disons un système de λ points A (l fois). Il y aura dans ce cas, entre les points P', A , une correspondance $(0, \lambda)$, et les points unis du système sont

les points A mêmes; le nombre des points unis est donc λ ; les deux côtés de l'équation contiendront les termes égaux $l\lambda$ et $l(0 + \lambda)$ respectivement, qui se détruisent, ce qui fait voir qu'il est permis de négliger les points fixes A , et ne faire attention qu'aux points d'intersection variables.

Il est assez remarquable que le théorème général peut s'écrire sous cette forme plus simple

$$pa + qb + \dots = p(\alpha + \alpha') + q(\beta + \beta') + \dots,$$

en comprenant parmi les systèmes formés par les intersections des courbes $U = 0, \Theta = 0$, le système du point P' (k fois), et en posant pour ce système

$$a = 0, \quad \alpha = \alpha' = D;$$

le système du point P' (k fois) donne ainsi un terme $= 0$ au côté gauche, un terme $= 2kD$ au côté droit de l'équation.

Comme dernier exemple appartenant à la formule simple

$$a = \alpha + \alpha' + 2kD,$$

je prends:

4°. *Recherche du nombre des points sextactiques*, c'est-à-dire des points qui sont tels, que par chacun passe une conique qui a dans ce point un contact du cinquième ordre avec la courbe.—Il faut prendre pour les points P les intersections avec la courbe de la conique qui a au point P' un contact du quatrième ordre; les points unis seront ceux dont il s'agit. La courbe $\Theta = 0$ est la conique qui a au point P' un contact du quatrième ordre. On a ainsi, parmi les intersections, le point P' 5 fois; donc $k = 5$. À chaque point P' correspondent $2m - 5$ points P ; à chaque point P , $(10m^2 - 20m - 5 - 20\delta)$ points P' (j'emprunte le terme -20δ d'une formule que vient de donner M. Zeuthen); donc la formule donne pour le nombre des points unis

$$10m^2 - 18m - 10 - 20\delta + 10D,$$

c'est-à-dire

$$15m^2 - 33m - 30\delta.$$

Mais cette expression comprend le nombre $3m(m - 2) - 6\delta$ des inflexions; en effet, pour un point d'inflexion, la conique avec contact du quatrième ordre se réduit à la tangente prise deux fois, ce qui est une conique avec contact du cinquième ordre. Donc enfin le nombre des points sextactiques sera

$$m(12m - 27) - 24\delta,$$

ou, pour une courbe sans points doubles,

$$m(12m - 27);$$

ce qui s'accorde avec la valeur que j'ai trouvée par d'autres moyens, [341].

378.

REPORT OF A COMMITTEE APPOINTED BY THE BRITISH ASSOCIATION FOR THE ADVANCEMENT OF SCIENCE, TO CONSIDER THE FORMATION OF A CATALOGUE OF PHILOSOPHICAL MEMOIRS.

[From the *Report of the British Association for the Advancement of Science*, (1856), pp. 463–464.]

The Committee were appointed—on the occasion of a communication from Professor Henry of Washington, containing a proposal for the publication of Philosophical Memoirs scattered throughout the Transactions of Societies in Europe and America, with the offer of cooperation on the part of the Smithsonian Institute, to the extent of preparing and publishing, in accordance with the general plan which might be adopted by the British Association, a Catalogue of all the American Memoirs on Physical Science—to consider the best system of arrangement, and to report thereon to the Council.

The Committee are desirous of expressing their sense of the great importance and increasing need of such a Catalogue.

They understand the proposal of the Smithsonian Institute to be, that a separate Catalogue should be prepared and published for America.

In the opinion of the Committee,

The Catalogue should embrace the Mathematical and Physical Sciences, but should exclude Natural History and Physiology, Geology, Mineralogy, and Chemistry, which would properly form the subject-matter of a distinct Catalogue or Catalogues. The difficulty of drawing the line would perhaps be greatest with regard to Chemistry and Geology; but the Committee would admit into the Catalogue memoirs not purely Chemical or Geological, but having a direct bearing upon the subjects of the Catalogue.

The Catalogue should not be restricted to memoirs in Transactions of Societies, but should comprise also memoirs in the Proceedings of Societies, in Mathematical and Scientific Journals, in Ephemerides and volumes of Observations, and in other collections not coming under any of the preceding heads. The Catalogue would not comprise separate works.

The Catalogue should begin from the year 1800.

There should be a Catalogue according to the names of authors, and also a Catalogue according to subjects; the title of the memoir, date, and other particulars to be in each case given in full, so as to avoid the necessity of a reference from the one Catalogue to the other.

The Catalogue should, in referring to a memoir, give the number as well of the last as of the first page, so as to show the length of the memoir.

The Catalogue should give in every case the date of a memoir (the year only), namely, in the case of memoirs published in the Transactions of a Society, the date of reading, and in other cases the date on the title-page of the volume. Such date should be inserted as a distinct fact, even in the case of a volume of transactions referred to by its date.

The Catalogue should contain a list of volumes indexed, showing the complete title; in the case of transactions, the year to which the volume belongs, and the year of publication; and in other cases, the year of publication, and the abbreviated reference to the work.

The references to works should be given in a form sufficiently full to be easily intelligible without turning to the explanation of such reference.

The author's name and the date should be printed in a distinctive type, so as to be conspicuous at first sight; and generally the typographical execution should be such as to facilitate as much as possible the use of the Catalogue.

As to the Catalogue according to the authors' names, the memoirs of the same author should be arranged according to their dates.

As to the Catalogue according to subjects, the question of the arrangement is one of very great difficulty. It appears to the Committee that the scheme of arrangement cannot be fixed upon according to any *à priori* classification of subjects, but must be determined after some progress has been made in the preliminary work of collecting the titles of the memoirs to be catalogued. The value of this part of the Catalogue will materially depend upon the selection of a proper principle of arrangement, and the care and accuracy with which such principle is carried out. The arrangement of the memoirs in the ultimate subdivisions should be according to their dates.

The most convenient method of making the Catalogue would appear to be, that each volume to be indexed should be gone through separately, and a list formed of all the memoirs which come within the plan of the proposed Catalogue. Such list should be in triplicate, one copy for reference, a second copy to be cut up and arranged for the Catalogue according to authors' names, and another copy to be cut up and arranged for the Catalogue according to subjects.

The Committee have endeavoured to form an estimate of the space which the Catalogue would occupy. The number of papers in a volume of transactions is in general small, but there are works, such as the *Comptes Rendus*, the *Astronomische Nachrichten*, the *Philosophical Magazine*, etc., containing a very great number of papers, the titles of which would consequently occupy a considerable space in the Catalogue. Upon the whole, the Committee consider that, excluding America, they may estimate the number of papers to be entered at 125,000; or since each paper would be entered twice, the number of entries would be 250,000. The number of entries that could conveniently be brought into a page 4to. (double columns) would be about 30, so that, according to the above estimate, the Catalogue would occupy ten quarto volumes of rather more than 800 pages each.

It appears to the Committee that there should be paid Editors, who should be familiar with the several great branches respectively of the Sciences to which the Catalogue relates; but that the general scheme of

arrangement and details of the Catalogue should be agreed upon between all the Editors, and that they should be jointly responsible for the execution. It would of course be necessary that the Editors should have the assistance of an adequate staff of clerks.

The principal scientific transactions and works would be accessible in England at the Library of the British Museum, and the libraries of the Royal Society and other Philosophical Societies. It would be the duty of the Editors to ascertain all the different works which ought to be catalogued, and to procure information as to the contents of such of them as may not happen to be accessible.

The Catalogue according to authors' names would be the most readily executed, and this Catalogue, if it should be found convenient, might be first published. The time of bringing out the two Catalogues would of course depend upon the sufficiency of the assistance at the command of the Editors; but if the Catalogue be undertaken, it is desirable that the arrangement should be such, that the complete work might be brought out within a period not exceeding three years.

A. Cayley.
R. Grant.
G. G. Stokes.

13th June, 1856.

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NOTICES OF COMMUNICATIONS TO THE BRITISH ASSOCIATION FOR THE ADVANCEMENT OF SCIENCE.

[From the *Reports of the British Association for the Advancement of Science*, 1854 to 1860, *Notices and Abstracts of miscellaneous Communications to the Sections*.]

1. *On the Solution of Cubic and Biquadratic Equations*. Report, 1854, p. 1.
2. *On the Porism of the In-and-circumscribed Triangle*. Report, 1855, p. 1.

The porism of the in-and-circumscribed triangle in its most general form relates to a triangle the angles of which lie in fixed curves, and the sides of which touch fixed curves, but at present I consider only the case in which the angles lie in one and the same fixed curve which for greater simplicity I consider to be a conic. We have therefore a triangle ABC the angles of which lie in a fixed conic $\mathfrak{S}$ and the sides of which touch the fixed curves $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$. And if we consider the conic $\mathfrak{S}$ and the curves $\mathfrak{A}, \mathfrak{B}$ as given, the curve $\mathfrak{C}$ will be the envelope of the side AB of the triangle. Suppose that the curves $\mathfrak{A}, \mathfrak{B}$ are of the classes m, n respectively; there is no difficulty in showing that the curve $\mathfrak{C}$ is of the class $2mn$. But the curve $\mathfrak{C}$ has in general double tangents forming two distinct groups, the first group arising from the quadrilaterals inscribed in the conic $\mathfrak{S}$ and such that two opposite sides touch the curve $\mathfrak{A}$, and the other two opposite sides the curve $\mathfrak{B}$; the second group arising from quadrilaterals such that two adjacent sides touch the curve $\mathfrak{A}$ and the other two adjacent sides touch the curve $\mathfrak{B}$. The number of double tangents of the first group is $= mn(mn - 1)$, and the number of double tangents of the second group is $= mn(mn - m - n + 1)$; the number of double tangents of the two groups is therefore $= mn(2mn - m - n)$. The curve $\mathfrak{C}$ has not in general any inflexions, hence, being of the class $2mn$ with $mn(2mn - m - n)$ double tangents, it will be of the order $2mn(m + n - 1)$.

When the curves $\mathfrak{A}$ and $\mathfrak{B}$ are conics, the curve $\mathfrak{C}$ is therefore of the class 8, with 16 double tangents but no inflexions, consequently of the order 24. But there are two remarkable cases in which the order is further diminished.

First when each of the conics $\mathfrak{A}, \mathfrak{B}$ has double contact with the conic $\mathfrak{S}$. The four points of contact give rise to 8 new double tangents or there are in all 24 double tangents, the curve $\mathfrak{C}$ is therefore of the degree 8: and being of the class 8 with 24 double tangents, it must of necessity break up into 4 curves each of the class 2, i.e. into 4 conics. Each of these has double contact with the conic $\mathfrak{S}$, or attending to only one of the four conics we have the well-known theorem which I call the porism (homographic) of the in-and-circumscribed triangle, viz. "there are an infinity of triangles inscribed each in a conic, and such that the sides touch conics having each of them double contact with the circumscribed conic."

Secondly, the conics $\mathfrak{A}$ and $\mathfrak{B}$ may intersect the conic $\mathfrak{S}$ in the same four points. Here every tangent of the curve $\mathfrak{C}$ is in fact a double tangent belonging to the first-mentioned group, the curve $\mathfrak{C}$ in fact

consists of two coincident curves: each of them is therefore of the class 4. But this curve of the class 4 has itself four double tangents arising from the common points of intersection of the conics $\mathfrak{A}, \mathfrak{B}$ with the conic $\mathfrak{C}$; it must therefore break up into two curves each of the class 2, i.e. into two conics: each of these intersects the conic $\mathfrak{C}$ in the same four points in which it is intersected by the conics $\mathfrak{A}, \mathfrak{B}$. Attending only to one of the two conics we have the other well-known theorem which I call the porism (allographic) of the in-and-circumscribed triangle, viz. “there exist an infinity of triangles inscribed in a conic, and such that the sides touch conics, each of them meeting the circumscribed conic in the same four points.”

3. *On the Notion of Distance in Analytical Geometry.* Report, 1858, p. 3.

The author remarks that the principles of Modern Geometry show that any metrical property whatever is really based upon a purely descriptive property, and that these principles contain in fact a theory of distance—but that such theory has not been disengaged from its applications and stated in a distinct and explicit form. The paper contains an account of the theory in question, viz. it is shown that in any system of geometry of two dimensions the notion of distance can be arrived at from descriptive principles by means of a conic called the Absolute, and which in ordinary geometry degenerates into a pair of points.

4. *On Curves of the Fourth Order having Three Double Points.* Report, 1860, p. 4.

The paper is a short notice only of researches which the Author is engaged in with reference to curves of the fourth order having three double points. A curve of the kind in question is derived from a conic by the well-known transformation of substituting for the original trilinear coordinates their reciprocals: and the species of the curve of the fourth order depends on the position of the conic with respect to the fundamental triangle.

5. *On Curves of the Third Order.* Report, 1861, p. 2.

A curve of the third order or cubic curve is a section of a cubic cone and such cone is intersected by a concentric sphere in a spherical cubic. It is an obvious consequence of a theorem of Sir Isaac Newton’s that there are five principal kinds of cubic cones, or what is the same thing five principal kinds of spherical cubics—but the nature of these five kinds of spherical cubics was first distinctly explained by Möbius. They may be designated the simplex, the complex, the crunodal, the acnodal and the cuspidal: where crunode, acnode, denote respectively the two species of double points (nodes), viz. the double point with two real branches, and the conjugate or isolated point. The foregoing results are known: the special object of the paper is to establish a subdivision of the simplex kind of spherical cubics. The simplex kind is a continuous reentering curve cutting a great circle, to fix the ideas say the equator, in three pairs of opposite points, which are the three real inflexions of the curve. The three great circles which are the tangents at the inflexions and the equator divide the entire surface of the sphere into fourteen regions whereof eight are trilateral and the remaining six are quadrilateral. The curve may be entirely in six out of the eight trilateral regions, and it is in this case said to be simplex trilateral; or it may lie entirely in the six quadrilateral regions, and it is in this case said to be simplex quadrilateral; and there is an intermediate form, the simplex neutral; viz. in this case the three great circles tangents at the inflexions meet in a pair of opposite points and there are in all only twelve regions all of them trilateral; the curve lies entirely in six of these regions.

6. *On a Certain Curve of the Fourth Order.* Report, 1862, p. 3.

The curve in question is the locus of the centres of the conics which pass through three given points and touch a given line; if the equations of the sides of the triangle formed by the three points are $x = 0, y = 0, z = 0$, these coordinates being such that $x + y + z = 0$ is the equation of the line infinity, and if $\alpha x + \beta y + \gamma z = 0$ be the equation of the given line, then (as is known) the equation of the curve is

$$\sqrt{\alpha x(y + z - x)} + \sqrt{\beta y(z + x - y)} + \sqrt{\gamma z(x + y - z)} = 0.$$

The special object of the communication was to exhibit the form of the curve in the case where the line cuts the triangle, and to point out the correspondence of the positions of the centre upon the curve, and the point of contact on the given line.

7. *On the Representation of a Curve in Space by means of a Cone and Monoid Surface.* Report, 1862, p. 3.

The author gave a short account of his researches recently published in the *Comptes Rendus*. The difficulty as to the representation of a curve in space is, that such a curve is not in general the complete intersection of two surfaces; any two surfaces passing through the curve intersect not only in the curve itself, but in a certain companion curve, which cannot be got rid of; this companion curve is in the proposed mode of representation reduced to the simplest form, viz. that of a system of lines passing through one and the same point. The two surfaces employed for the representation of a curve of the n th order are, a cone of the n th order having for its vertex an arbitrary point (say the point $x = 0, y = 0, z = 0$), and a monoid surface with the same vertex, viz. a surface the equation whereof is of the form $Qw - P = 0$, P and Q being homogeneous functions of (x, y, z) of the degrees p and $p - 1$ respectively (where p is at most $= n - 1$). The monoid surface contains upon it $p(p - 1)$ lines given by the equations ($P = 0, Q = 0$); and, the cone passing through $n(p - 1)$ of these lines (if, as above supposed, $p < n - 1$, this implies that some of these lines are multiple lines of the cone), the monoid surface will besides intersect the cone in a curve of the n th order.

8. *On a Formula of M. Chasles relating to the Contact of Conics.* Report, 1864, p. 1.

The author gave an account of the recent investigations of M. Chasles in relation to the theory of conics, viz., M. Chasles has found that the properties of a system of conics, containing one arbitrary parameter, depend upon two quantities called by him the characteristics of the system; these are, μ , the number of conics of the system which pass through a given point, and, ν , the number of conics of the system which touch a given line; or, say, μ is the parametric order, ν the parametric class, of the system. And he exhibited a transformation obtained by him of a formula of M. Chasles for the number of conics which touch five given curves, viz., if (M, m) (N, n) (P, p) (Q, q) (R, r) be the orders and classes of the five given curves respectively, then the number of curves is

$$= (1, 2, 4, 4, 2, 1)(M, m)(N, n)(P, p)(Q, q)(R, r),$$

where the notation stands for $1 \cdot MNPQR + 2\Sigma mNPQR + 4\Sigma mnPQR + \text{etc.}$ The transformed formula in question was communicated by the author to M. Chasles, and had appeared in the *Comptes Rendus*; but it is, in fact, included in a very beautiful and general theorem given in the same Number by M. Chasles himself.

[End of Vol. V.]